Annex 4-5

Section 4: Members of the consortium

4.1. Participants (applicants)

4.1.1. UNIPI

Università di Pisa - UNIPI	
www.unipi.it	
Italy	*University*

Description

The University of Pisa, founded in 1343, is one of the most ancient and prestigious universities in Europe. UNIPI counts twenty large departments covering all disciplinary areas, with high level research centres in the sectors of physics, computer science, engineering, mathematics, among others. The academic staff counts around 1400 professors and researchers, 1500 administrative employees, and more than 50.000 students (including students from other Italian regions and from abroad). The vitality of higher education and scientific research determines its strength: UNIPI is committed to promoting and supporting research in every field of knowledge, encouraging innovation and openness to new subjects and collaborations between different disciplines. Thanks to the quality of the research undertaken by its academic staff, individually or in teams, UNIPI holds a prominent position in the national and international scientific context; the 2019 QS World University Rankings by Subject ranks UNIPI in the top 100 among 1200 Universities in five subjects.

The Department of Civil and Industrial Engineering of the University of Pisa (DICI-UNIPI) is located in via Diotisalvi, 2, in the centre of Pisa. The Multifunctional Bio-Ecocompatible Materials Laboratory, coordinating CEM-WAVE project activities, consists of about 15 researchers and PhD students and 3 technicians and administrative staff. The Department was established in 2012, following a major reform of the Italian higher education system, as a merger of the former departments of Aerospace Engineering, Chemical Engineering, Industrial Chemistry and Materials Science, Mechanical, Nuclear and Production Engineering, and a part of the department of Civil Engineering. The departmental research covers a wide variety of skills in Materials Science: characterization and development of materials, polymer processing, design and modelling of mechanical properties. These research activities have a strong influence in areas of high social and economic impact, such as materials and environment, health, energy, industrial. The research activity is carried out in the DICI-UNIPI laboratories by employing advanced scientific instruments in collaboration with a variety of both national and international academic and private institutions.

Main tasks

UNIPI will coordinate the CEM-WAVE project. UNIPI will be involved in all work packages (WP1-WP10), with different commitments.

The scientific WP with the higher UNIPI involvement is WP3 - Optimization of MW-CVI process for non-oxide and oxide CMC's production (WP leader), where UNIPI will lead two tasks related to the non-oxide CMCs (3.1 and 3.4). In more detail, UNIPI will perform research and tuning of the operating parameters for MW-CVI non-Oxide CMC's production and produce the infiltrated non-oxide CMCs samples.

Being the coordinator, UNIPI will lead the WP10 – Coordination and management and all its tasks.

UNIPI will also collaborate, in WP2, with IPCF-CNR for the modelling of reactor cavity in which the implementation of solid state technology will be updated. Furthermore, UNIPI will support in all WP in which the knowledge about the non-oxide of MW-CVI production is required.

Relevant publications/products/services/activities

Projects:

- HELM (FP7-280464) – "High-frequency ELectro-Magnetic technologies for advanced processing of ceramic matrix composites and graphite expansion".
- NANOCATHEDRAL (H2020- 646178) – Innovation for Europe Cultural Heritage Protection and Conservation.
- EVOLUTION (FP7-314744): The Electric Vehicle revOLUTION enabled by advanced materials highly hybridized into lightweight components for easy integration and dismantling providing a reduced life cycle cost logic.
- ECOAT (H2020-837863): ECO sustainable multifunctional bio-based COATings with enhanced performance and end of life options.
- BIONTOP (H2020-837761): Novel packaging films and textiles with tailored end of life and performance based on bio-based copolymers and coatings.

Publications:

1. Lazzeri, A. CVI Processing of Ceramic Matrix Composites. Ceram. Compos. Process. Methods 2012, 313–349.
2. Coltelli, M.-B.; Lazzeri, A. Chemical vapour infiltration of composites and their applications. Chem. Vap. Depos. Adv. Technol. Appl. 2019, 363.
3. Cioni, B.; Lazzeri, A. Modeling and Development of a Microwave Heated Pilot Plant for the Production of SiC-Based Ceramic Matrix Composites. Int. J. Chem. React. Eng. 2008, 6.
4. Miceli, V.; Cioni, B.; A., L. Microwave Processing of Liquid Phase Sintered Alumina. In Materials Science and Technology; 2009.
5. Genovesi, S.; Costa, F.; Cioni, B.; Miceli, V.; Gallone, G.; Levita, G.; Lazzeri, A.; Annino, G.; Monorchio, A.; Manara, G. Zirconium tin titanate (ZST) for miniaturized high impedance surfaces: Microwave dielectric properties and applications. In 2009 IEEE Antennas and Propagation Society International Symposium; 2009; pp. 1–4.

Products / Services / activities*:*
N/A

Key personnel

Prof. Eng. ANDREA LAZZERI (male), project coordinator, is Full Professor of Materials Science and Engineering. He is specialized in preparation and use of micro-and nano-particles in polymeric composites, rubber toughening of plastics, deformation and fracture of materials and in microwave chemical vapor infiltration (MW-CVI) processing of ceramic matrix composites (CMCs). He is the author of over 172 publications, including 78 works in national and international journals (3508 Citations, h-index 29, i-20 62) and he is co-author of 6 patents. He is the Head of the Interdepartmental Center for Materials Science and Engineering of the University of Pisa and head of MSc degree course in Materials and Nanotechnology. He coordinated three European projects on Materials Science, in particular project HELM aimed at making production technologies more efficient for the production of CMCs through the use of heating technologies that use high frequency electromagnetic fields.

Prof. MARIA BEATRICE COLTELLI (female) Associate Professor with 15 years of experience on materials characterization and study of the correlations between molecular structure, morphology and properties of polymers, biopolymers, blends and composites, especially related to their relation with environment, and CVI production technologies of CMCs. She participated in research activities during HELM project focusing in particular on the study and prediction of deposition rates in the MW-CVI process and morphological analysis of infiltrated samples. She is author and co-author of 40 publications in international journals and two patents. She followed the scientific activity of European projects on Materials Science and Nano-technology. Currently she supports the didactical activity of courses for Chemical Engineers and Construction Engineers.

Prof. PATRIZIA CINELLI (female) Associate Professor with a PhD in Chemistry at Pisa University partly performed at the United States Department of Agriculture (USDA), Peoria, IL, USA. 20 years of experience in materials science, polymer processing and characterization, biodegradation and life cycle assessment. She is co-

author of over 55 papers in peer-reviewed journal, 9 book chapters, and 8 patents on innovative materials. She is associate professor of Pisa University for Applied Chemistry and Materials Technology at Master Degree student of Construction Engineering and Architecture at Pisa University, and Master Degree students of Materials and Nanotechnology. She has worked in over 15 EC project from FP5, FP6, FP7 and Horizon 2020, directly following writing, scientific activity, and reporting.

Dr. FRANCESCA SIGNORI (female) is Assistant Professor in Materials Science and Technology at the University of Pisa. PhD in Biomaterials at University of Trento, with part of the project performed at the University of Twente, The Netherlands, focused in chemical synthesis and modification of biodegradable polymers and biopolymers. She has 15 years of experience in Materials Science, Polymer and Ceramic Processing and Characterization. She is co-author of 33 papers in peer reviewed journal, 1 book chapter, and 1 patent on innovative materials.

Dr. Eng. LAURA ALIOTTA (female) She is a Chemical Engineer with a PhD in Chemical and Materials Engineering at Pisa University. She discussed the PhD thesis on 5 July 2019 with the title: "Development of advanced composites with tailored strength and toughness". She works mainly on the evaluation and modulation of the thermo-mechanical properties of polymeric and ceramic composites materials. She studies the development of the mechanical and micromechanical properties of composite materials through the use of analytical models. She collaborated in HELM project for the silicon carbide based ceramic matrix composite production in the pilot scale plant located at CNR (Pisa).

Dr. Eng. VITO GIGANTE (male). He is a Chemical Engineer with a PhD in Chemical and Materials Engineering at Pisa University. He discussed the PhD thesis on 5 July 2019 with the title: "Processing and characterization of PLA/PBAT blends for toughened rigid and flexible items". He works mainly on fracture and impact mechanics of polymeric and ceramic composites. Furthermore, he works on processability and physical properties of multiphase composite systems for different applications. He collaborated in HELM project for the silicon carbide based ceramic matrix production in the pilot scale plant located at CNR (Pisa).

Eng. ROBERTO D'AMBROSIO (male). He is a Chemical Engineer graduated in 2017 with a master degree thesis on "Production of Ceramic Matrix Composites (CMCs) through Chemical Vapor Infiltration". He's currently a PhD Student at the Department of Civil and Industrial Engineering with topic related to the "Production of CMCs using non conventional energy sources". His research activity is mainly focused on the production, study and modeling of silicon carbide based CMCs through Microwave assisted Chemical Vapor Infiltration process (MW-CVI), carrying out the experimental tests at the pilot plant designed and developed during the HELM project.

<u>Relevant infrastructure</u>

- Pilot plant for MW-CVI process equipped with:
    - Furnace (Stainless steel vessel with inner graphite chamber)
    - x3 Magnetrons SAIREM Iberica, 3 kW available each (total of 9 kW) operating around 2.45 GHz
    - Liquid ring pump Tuthill KLRC75
    - Booster pump Tuthill KMBD400
    - Brazed plate heat-exchanger Thermex
    - H_2 sensor Sicor CP1 Plus IL
    - Pressure control valve MKS T3bi
    - Water-cooling system Daikin EWAQ021BAWP
    - Digital pyrometer Dias Pyrospot DSR 10NV
    - PH-meter Process instruments CRONOS
    - Microwave leakage meter SAIREM IFP 2450 MHz
    - Earthed Glass Fibre Tapes Electrothermal HT95503
- Nabertherm LHT02/17 high temperature oven
- MTS Criterion model C43 Universal Testing Machine
- Video extensometer (Genie HM1024 Teledyne DALSA camera) interfaced with a ProVysis software (Fundamental Video Extensometer)
- MTS climate chamber for quasi-static tests (Thermcraft Incorporated model LBO-24-10-1T-J14641/1A)

- INSTRON 5500r Universal Testing Machine
- Scanning electron microscopy (SEM) FEI Quanta 450 FEG
- Dynamic mechanical thermal analysis (DMTA) Gabo Eplexor® 100N
- FT-IR Nicolet 380 spectrometer
- Optical microscope Olympus BX60

4.1.2. IPCF-CNR

Istituto per i Processi Chimico-Fisici, Consiglio Nazionale delle Ricerche (IPCF-CNR)	
www.ipcf.cnr.it, www.cnr.it	
Italy	*Public Research Institution*

Description

The National Research Council (CNR) is the largest public research institution in Italy, the only one under the Research Ministry performing multidisciplinary activities.

Founded as legal person on 18 November 1923, CNR's mission is to perform research in its own Institutes, to promote innovation and competitiveness of the national industrial system, to promote the internationalization of the national research system, to provide technologies and solutions to emerging public and private needs, to advice Government and other public bodies, and to contribute to the qualification of human resources.

In the Cnr's research world, the main resource is the available knowledge which means people, with their skills, commitment and ideas. This capital comprises more than 8.000 employees, of whom more than half are researchers and technologists.

At IPCF, the research activity is intrinsically interdisciplinary and focused on Chemical and Physical topics, as soft matter, disordered systems, interfaces, composites and nanostructured materials, supramolecular architecture and biophysics, complex fluids and colloids. Main interests concern with the investigation of phenomena as self-organization, surface interactions, relaxation and transport properties, from nano- to meso- scale. IPCF is involved in the design and manufacturing of tailored materials, on theoretical models, on computational and analytical techniques and in the development of new devices and experimental methodologies.

Some of the activities are highly relevant for the society (energy, environment, health): development of dye sensitized solar cells, pollutant sensors and techniques for environmental clean-up, development of technologies for industry and nanomedicine.

IPCF has a long term experience in the development of microwave devices and techniques, and in their application to the characterization and processing of materials. In this context, relevant international projects are given by the NATO Science for Peace project SfP 980881 'Tantalum-Free Microwave Dielectric Resonators with Enhanced Quality Factor' (2005-2008), and by the FP7 large scale EC project HELM 'High-frequency electromagnetic technologies for advanced processing of ceramic matrix composites and graphite expansion' (2012-2016).

Main tasks

IPCF-CNR will lead WP2 and give an essential contribution to WP3.

WP2 will focus on the high temperature dielectric characterization of selected non-oxide and oxide fibers (Task 2.1), on the rigorous numerical modeling of the EM - Heating problem inside MW-CVI reactors (Task 2.2), and on the set-up of MW-CVI reactors fed by solid state sources (Task 2.3).

Task 2.1 and Task 2.2 will be led by IPCF, on the basis of the experience gained as leader of WP4 of the EC project HELM ('High-frequency electromagnetic technologies for advanced processing of ceramic matrix composites and graphite expansion'), dedicated to the 'Development of new methodologies for high temperature microwave/radiofrequencies processing and process monitoring'.

Relevant publications/products/services/activities

Projects:

- FP7 large scale EC project HELM 'High-frequency electromagnetic technologies for advanced processing of ceramic matrix composites and graphite expansion' (2012-2016).

Publications:

- Bianchi G, Vavassori P, Brais V, Annino G, Nagliati M, Mallah M, Gianella S, Valli M, Orlandi M, Ortona A (2015). Reactive silicon infiltration of carbon bonded preforms embedded in powder field modifiers heated by microwaves. CERAMICS INTERNATIONAL, vol. 41, p. 12439-12446, ISSN: 0272-8842
- Bussolino G C, Annino G, Ferrari C, Righini F (2011). Virtual Experiments by Pulse Heating Techniques: Cylindrical Tungsten Specimens. INTERNATIONAL JOURNAL OF THERMOPHYSICS, vol. 32, p. 2716-2726, ISSN: 0195-928X
- Shimada T, Ichikawa K, Minemura T, Kolodiazhnyi T, Breeze J, McN Alford N, Annino G (2010). Temperature and frequency dependence of dielectric loss of Ba(Mg1/3Ta2/3)O3 microwave ceramics. JOURNAL OF THE EUROPEAN CERAMIC SOCIETY, vol. 30, p. 331-334, ISSN: 0955-2219
- Kolodiazhnyi T, Annino G, Spreitzer M, Taniguchi T, Panariello A, Fitzpatrick W (2009). Development of Al2O3 - TiO2 composite ceramics for high-power millimeter-wave applications. ACTA MATERIALIA, vol. 57, p. 3402-3409, ISSN: 1359-6454
- Kolodiazhnyi T, Annino G, Shimada T (2005). Intrinsic limit of dielectric loss in several Ba(B'1/3B''2/3)O3 ceramics revealed by the whispering-gallery mode technique. APPLIED PHYSICS LETTERS, vol. 87, 212908, ISSN: 1077-3118

***Products / Services / activities**:*

- Microwave dielectric characterization of samples at temperatures variable from room T to 1200 °C
- Design of MW reactors by means of rigorous multiphysics modeling

Key personnel

G. Annino, researcher, WP2 and Tasks 2.1 and 2.2 leader, expert in electromagnetic modeling, microwave dielectric characterization, and electromagnetic field-matter interaction. Master of Sciences in Physics (cum Laude). Author or co-author of more than 50 publications on international journals, most of them on electromagnetic devices and techniques, and on their application to the characterization and processing of materials, including the recent 'Electric field modulation of magnetic exchange in molecular helices', Nature Materials, vol. 18, p. 329 (2019). Author of 2 international patent applications, 'Hyperpolarization and Multiple Irradiation Probe Head for Magnetic Resonance Techniques' and 'Structure of an NMR probe head'. Leader of the work package 'Development of new methodologies for high temperature microwave/radiofrequencies processing and process monitoring' of the Large Scale EC project 'High-frequency electromagnetic technologies for advanced processing of ceramic matrix composites and graphite expansion' (HELM). Outstanding reviewer for the journals IEEE Transactions on Instrumentation and Measurement (2010), Review of Scientific Instruments (2012), and Journal of Magnetic Resonance (2016).

A. Cintio, research fellow, expert in multiphysics modeling, dielectric characterization and analysis, electromagnetic phenomena. Master of Sciences in Physics (cum Laude). Author or co-author of 4 papers in international journals and 5 contributions to international conferences, including the 'Design of a Pilot-scale Microwave Heated Chemical Vapor Infiltration Plant: An Innovative Approach' (10th International Conference on High Temperature Ceramic Matrix Composites, Bordeaux (F), 22-26 September 2019), 'Design of an Overmoded Resonant Cavity-Based Reactor for SiC-Based Ceramic Matrix Composites Production' (17th International Conference on Microwave and High Frequency Heating AMPERE 2019, 9-12 September 2019), and 'Dielectric Characterization of Si, SiC, SiC/SiC, and C/SiC Samples' (17th International Conference on Microwave and High Frequency Heating AMPERE 2019, 9-12 September 2019).

Relevant infrastructure

- Setup for the microwave dielectric characterization of samples at temperatures from room T to 1200 °C
- High-Performance Computing cluster equipped with 4 computational nodes, 64 cpu, and 1 TB RAM
- Finite-element software Comsol Multiphysics
- 4-ports Agilent vector network analyzer covering frequencies from 10 MHz to 43.5 GHz

4.1.3. FM

Fricke und Mallah Microwave Technology GmbH		
https://www.microwaveheating.net		
Germany	*GmbH*	

Description

The company Fricke und Mallah Microwave Technology GmbH in Peine / Lower Saxony has been a leading manufacturer of microwave heating and drying systems as well as microwave components (generators, coupling systems, RF measurement technology) since its foundation in 1995. We are located in Peine, Lower Saxony. We have developed and built microwave rotary kilns and continuous furnaces for the heat treatment of various products under different atmospheres (normal atmosphere, vacuum, overpressure). Due to the physical facts, microwave radiation can penetrate deeply into numerous products and since this penetration depth is dynamic in time due to the selectivity of microwave heating, microwave drying is homogeneous, gentle and fast.

We develop and manufacture high-quality microwave systems for the wood, ceramic and life industries up to 1 MW microwave power!

As high voltage power supplies we have developed and manufactured innovative switching power supplies up to 6 kW. The switching power supplies (inverter technology) are characterized by low weight and smaller dimensions than conventional high-voltage power supplies (transformer technology). In addition, they allow a more precise power setting and cause a longer life of the magnetrons!

For 5 years we have been developing and manufacturing very compact, intelligent and inexpensive industrial semiconductor-based microwave generators with 915 MHz and 2.45 GHz, where frequency and phase adjustment as well as automatic power adjustment are possible. This semiconductor-based microwave technology is very promising because the problems of homogeneity of the microwave field and power matching, which are known and automatically arising in magnetron technology, can be solved without great technical effort. This significantly increases the efficiency of converting microwave energy into thermal energy in the heating or drying chamber.

We have a very efficient simulation department for electromagnetics and thermodynamics, consisting of 4 experienced engineers with many years of professional experience. We use CST Studio Suite software for modelling microwave chambers, absorber zones and microwave coupling systems (antennas).

Our well-known customers include Siemens, Daimler, DLR (German Aerospace Center), Saint Gobain, VW, BASF, Lufthansa, Kraftfood, Nestle and Tönnies Fleisch as well as several universities such as Bayreuth, Duisburg, Rostock, Braunschweig, Dortmund and Erlangen.

Main tasks

Our main activities in the CEM-WAVE project are:

- Integration and Set-up of MW-CVI reactors fed by Solid State Sources at working frequencies 900…930 MHz and 2.4…2.5 GHz including the corresponding microwave measurement and power matching systems
- Support IPCF-CNR by modelling and simulation of the microwave fields

Relevant publications/products/services/activities

Projects:

- FP7 large scale EC project **HELM** ‘High-frequency electromagnetic technologies for advanced processing of ceramic matrix composites and graphite expansion’ (2012-2016).
- **SuperCarbon.** Carbonfaservliesen mit hierarchischer Oberflächenporosität und erhöhter Energiedichte für Superkondensatorelektroden (SuperCarbon). Teilvorhaben: Mikrowellen Umwandlung – Untersuchungen zu. Carbon fiber fleeces with hierarchical surface porosity and increased energy density for Super capacitor electrodes (SuperCarbon). Subproject: Microwave conversion - studies on Microwave-based PAN nonwoven stabilization and carbonization
- **Pyrolyse Rotary Tube.** Entwicklung eines Verfahrens zur ressourceneffizienten Recycling von Leiterplattenschrott mittels Mikrowellenpyrolyse in Drehrohröfen; Konstruktion und Fertigung eines

Demonstrators der Drehrohranlage zur Erprobung des Verfahrens. Development of a process for the resource-efficient recycling of printed circuit board scrap by means of microwave pyrolysis in rotary kilns; Design and manufacture of a demonstrator of the rotary kiln system to test the process

- **ReOrgAl** – Steigerung der Energie- und Ressourceneffizienz des Recyclings organik-kontaminierter Aluminiumschrotte; Teilprojekt: Planung und Bau einer kontinuierlichen Mikrowellenpyrolyse zur thermischen Vorbehandlung Aluminiumschrotten
  ReOrgAl - Increasing the energy and resource efficiency of recycling organically contaminated aluminum scrap; Subproject: Planning and construction of a continuous microwave pyrolysis for the thermal pretreatment of aluminum scrap
- **DEMETO.** DEMETO proposes the industrialization and demonstration at full-scale of a new industrial process which allows for the first time to recycle chemically PET bottles, food containers and even textiles in a highly profitable and environmentally sustainable way.
- **polynSPIRE.** Demonstration of Innovative Technologies towards a more efficient and sustainable plastic recycling

Publications:

- Photorec, Recovery of rare strategic metals from EOL thin film PV modules published in the Book: Innovative Technologies for Resource Efficiency, Strategic Metals and Minerals

- Reactive silicon infiltration of carbon bonded preforms embedded in powder field modifiers heated by microwaves
  - Giovanni Bianchi, Paolo Vavassori, BraisVila, Giuseppe Annino, Marco Nagliati, Marcel Mallah, SandroGianella, MassimilianoValle, MarcoOrlandi, AlbertoOrtona,
  - ICIMSI-SUPSI, Manno, Switzerland
  - Petroceramics SPA, Stezzano, Italy
  - Fricke und Mallah Microwave Technology GmbH, Peine, Germany
  - Istituto periProcessi Chimico-Fisici, UOS Pisa, Consiglio Nazionale delle Ricerche, Pisa, Italy
  - Brembo SGL Carbon Ceramic Brakes, Stezzano, Italy
  - Erbicol SA,Viale Carlo Pereda 22, 6828 Balerna, Switzerland

***Products / Services / activities**:*

- The company Fricke und Mallah Microwave Technology GmbH has more than 20 years experiments in the field of microwave heating. In the last 5 years the field of solid state devices for the generation of the RF power is successful used and consistent evolved. The solid state technology is used in the frequency range of 900 MHz up to 930 MHz as well as in the frequency range of 2400 MHz up to 2500 MHz. For the 915 MHz a generator with an output power of 1000 watt and for 2450 MHz a generator with an output power of 500 watt is used. Furthermore, systems up to 4000 watt (915 MHz) or 2000 watt (2450 MHz) a realized. Therefore, different power combiners are developed in as coaxial, waveguide or as combination (coaxial input and waveguide output) system. Furthermore, different measurement devices like a waveguide based directional coupler as well as power detectors was developed for monitoring and controlling of such high-power systems based on solid state technology.

- Own new international patent applications
  - Apparatus for generating a plasma beam in the MHz and GHz range with TEM and waveguide modes
  - Pasteurization of ready meals in hermetically sealed containers (COSTPANO®)
  - MICROWAVE-3D-SANDFORMBOX

<u>Key personnel</u>

Mr. Marcel Mallah, Dipl.-Eng. Study of High Frequency and Microwave Technology 1986-1992 at the technical University Braunschweig in Germany. 3 Years work experience in Switzerland (developing of microwave heating systems for resins at Ciba Geigy) and South Germany mainly developing high temperature microwave heating systems for Siemens (sintering ovens for atomic pellets) and microwave heating systems under different atmospheric conditions vacuum and high pressure for research applications. Since 1995 founder and CEO of FM

Mr. Probst, Steffen, Dr.-Ing. Studies (Master of Science), Leibniz University Hannover, Masters Degree: Electrical Engineering and Information Technology, specialization: Communications Engineering2004-2007 "Material Engineer" at Agilent Technologies Germany. PhD-Work at Leibniz Universität Hannover, Institute for High frequency technology and radio systems:

- Focus on independent research and development in the areas of high-efficiency Power amplifier and non-linear measurement technology
    - Measurement of high-frequency voltages and currents in power amplifiers
    - Development and construction of Doherty amplifiers in different frequency ranges to increase efficiency via the tax area
    - Investigation of the impact of load mismatch on the efficiency of high efficiency power amplifiers
    - Design of adaptive matching networks to compensate for load mismatch

Since 01.01.2019 Development Engineer at Fricke and Mallah Microwave Technology GmbH

Mr. Jerome Ogrzall, M. Sc. Study of Electrical Engineering 2005-2008 at the technical University RWTH Aachen in Germany. Study of Telecommunication Engineering 2009-2014 at the University of Applied Sciences Aachen (FH Aachen) in Germany. 2014-2016 Development Engineer High Frequency Technology - FH Aachen. Since November 2016 High Frequency Development Engineer at Fricke and Mallah Microwave Technology GmbH.

Relevant infrastructure

The company Fricke und Mallah Microwave Technology GmbH has a couple of RF measurement equipment to characterize passive as well as active RF components up to a power of 20000 watt. This included a vectorial network analyzer up to a Frequency of 8 GHz, for the measurement of e.g. directional coupler, power combiner, and filter. Furthermore, the vectorial network analyzer or the handheld version of the vectorial network analyzer can be used to measure the match and the decoupling of the antennas of a designed chamber. Nevertheless, for the characterization of the active components such the generator, signal source, or power amplifiers, different power detectors with a dynamic range of 80 dB as well as a spectrum analyzer up to a frequency of 13.6 GHz can be used. All measurement equipment can be used with coaxial or waveguide systems.

4.1.4. FHF

Fraunhofer ISC - Center for High Temperature Materials and Design	
www.htl.fraunhofer.de	
Germany	*Research facility*

Description

Fraunhofer ISC - Center for High Temperature Materials and Design is a research facility with the focus on ceramics and high temperature composites. The institute has the focus on two business areas: the high temperature characterization and ceramic matrix composites (CMC). Fraunhofer HTL offers large background know-how on the characterization and manufacturing of any kind of CMC. Key activities in the field of CMCs are the ceramic fibre manufacturing and fibre coating by using ceramic precursors. All fibres can be tailored to textiles by using adapted textile machines.

For the material design, Fraunhofer-Center HTL selects suitable starting materials - fibres and matrices - to obtain the desired material properties. Typical fibres for processing are carbon, aluminum oxide, mullite and silicon-carbide fibres. Ceramic fibres are developed inhouse from polymer precursors and manufactured up to pilot scale. The fibres are further processed as fibre rovings or textile fibre preforms. For non-oxide CMCs, the matrices preferably include carbon precursors and silicon polymers, which are converted to ceramics by high-temperature processes. In addition, carbide matrices are produced with the so-called Liquid Silicon Infiltration Process, shortened LSI process, by infiltrating a silicon melt into porous carbon preforms. With O-CMCs, typically matrices based on aqueous powder dispersions in the system Al2O3-SiO2-ZrO2 are used. Fraunhofer-Center HTL utilizes different material concepts (weak interface-, weak matrix- composites) to adjust the quasi-ductile properties of fibre composites. A closed process chain in laboratory scale is available for the production of samples for new fibre composites, often carried out as small-format plates or tubes. Fraunhofer-Center HTL develops reproducible manufacturing methods and statistically secures the measured material properties.

Fibre composite components are manufactured at Fraunhofer-Center HTL as prototypes or as small series in closed-loop processes. The basis for this is successful material development with selected matrices and textile fibre preforms, where the material properties are adjustable. Our development includes component manufacture in-line with fibre requirements as well as manufacturing. For the heat and high-temperature treatment of fibre-reinforced composite components, multiple heat treatment facilities as well as a specific high temperature furnace system up to 2400 °C are available. Components from fibre-reinforced ceramics are designed on the basis of finite element analyses.

For the production of fibre composite components, pilot plants are available at Fraunhofer-Center HTL that allow upscaling to large component dimensions up to 1000 mm. For the upscaling of newly developed ceramic fibres, a fibre pilot plant is under construction, which allows a production in ton scale. For the manufacturing of green bodies hot presses, a CNC-controlled winding system and a continuously operating prepreg system for the coating of fibre preforms are available. Other plants for textile fibre processing are situated at the application centre for Textile Fiber Ceramics TFK at the Münchberg facility.

The composites are consolidated by heat treatment up to about 200 °C and by high temperature treatment up to 2400 °C conversed into CMC. By means of mechanical processing of individual components and high temperature joining techniques, Fraunhofer-Center HTL can create very complex component geometries. Ceramic coatings enable use under corrosive environmental conditions and thus an increase in service life. An advanced 5-axis machining centre is available for the intermediate and finishing processes. When needed, Fraunhofer-Center HTL also offers a concept to small-series production of components after successful prototype manufacturing. The manufacturing can be undertaken customer-specifically by quality assurance measures. Hence, a transfer of results and technology into the production scale is possible.

At Fraunhofer-Center HTL, a series of non-destructive, mechanical and thermos-physical testing methods is available. The structure and homogeneity of materials can be analyzed with non-destructive testing (NDT) methods. Using NDT, material defects, which occur during manufacturing or whilst in use, can be detected. Mechanical tests are possible at room temperature and at temperatures up to approx. 1500 °C. A large number of standardized procedures (DIN, EN, ISO, ASTM) can be used. They can be realized with both quasi-static and dynamic-cycled stress profiles, so-called fatigue tests. Fibre preforms, samples and components can be analyzed before and after the examination with computed tomography and other analysis methods to detect damage or material changes.

Main tasks

In the CEM-WAVE project, Fraunhofer will be responsible for the WP1 as a leader and will also contribute to WP 6 with its coating activities.

WP1-task 1.2: Fibre coating on non-oxide fibres by a continuous wet-chemical process

WP1-task 1.3: Fibre coating on oxide fibres by a continuous wet-chemical process

WP1-task 1.4: 3D CMC preforms, both oxide and non-oxide, with adapted porosity for the microwave densification

WP6-task 6.1: Development of environmental barrier coatings EBC on non-oxide CMC

Relevant publications/products/services/activities

Projects:

- "Tra_Turb": CMC Support structures in the hot gas area of gas turbines, 01/2019-12/2021, BMBF funded German project (grant number: 03XP0169F)
- „Ker_Twk": Ceramic in gas turbines, 10/2018-03/2022, (grant number: 20T1724D)
- "CMC-SiC": Development of SiC/SiC material and technology, 10/2016-09/2019, funded by the Bavarian Ministry of Education (grant number: 43-6622/438/19)
- „CMC-Engine" - Lightweight structures of SiC/SiC-Ceramic for gas turbines of the next generation, 01/2017-06/2020, FHG funded program (grant number 833 886)
- "CaGeFa": Development of fiber coating and fiber preforms by using textile technology for efficient CMC´s, 10/2015-09/2018, project funded by the German Ministry of Research and Development (grant number: 03X0143C)

Publications:

- Maier, J.; Nöth, A.; Schönfeld, K.: BN-Based Fiber Coatings by Wet-Chemical Coating, Key Engineering Materials, Vol. 809, pp. 421-426, 2019.Eckardt, C.; Friedrich, M.; Wamser, T.; Wolff, N.; Metzger, K.: Faserverstärkte Keramik-Armierungen für den Rohrleitungsbau, Chemie Technik April/2019, S. 38-40.
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- Rüdinger, A.; Pritzkow, W.: Die Entwicklung oxidkeramischer Faserverbundwerkstoffe am Fraunhofer ISC / Zentrum HTL in Zusammenarbeit mit W.E.C. Pritzkow Spezialkeramik, Keram. Z. 65, 2013, [3] pp.166-169. (zum Abstract)
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- Nöth A., (2016), Application number: DE16169999.6, VERFAHREN UND ANLAGE ZUR HERSTELLUNG VON HOMOGEN MIT EINER KERAMIK BESCHICHTETEN FASERN, BESCHICHTETE FASERN UND VERWENDUNGEN HIERVON, EP 3 103 781 A1
- Nöth A., Vogt J, (2019), Application number: DE102019206940.1, Keramik mit Korrosionsschutzschicht, Verfahren zu deren Herstellung und deren Verwendung, patent pending
- Maier J., Nöth A., (2019), Application number: DE10 2019 215 938.9, Faserbeschichtung aus Bornitrid über kostengünstige, luftstabile und flüssige Precursoren, patent pending
- Raether F., Vogt J., Schmidt J (2019), Application number: DE102019202695.8, Verfahren zur Herstellung von Prepregs für die Herstellung faserverstärkter Keramikbauteile

***Products / Services / activities**:*

- Complete process chain for CMCs: form the ceramic fibre to the CMC component
- Prototype manufacturing up to 1,000 mm and joining
- High temperature testing by using international standards: destructive and non-destructive tests
- Thermo-optical measurements for the determination of thermo-physical properties
- Modelling of ceramics and composites at all scales from micro-to-macro

Key personnel

Jens Schmidt received his doctoral degree in Mineralogy at University Berlin in 1997 and is an expert for the manufacturing and testing of CMCs for more than 20 years. After a period as a researcher at Daimler Benz in the mid 90s, he conducted research projects at the German Aerospace Center DLR. He was a visiting scientist at the University of California Berkeley in 2006. Since 2010, he is a group leader at Fraunhofer ISC - Center for High Temperature Materials and Design. He has experience with R&D management and coordinated several national and European projects as a project leader. He published more than 25 patents and more than 30 scientific papers. He is familiar with all types of CMCs like $Al_2O_{3f}/Al2O_3$-ZrO_2, C_f/SiC and SiC_f/SiC. In recent years, he is focusing on the development of SiC/SiC by investigating a low cost and time efficient silicon infiltration process for gas turbine applications.

Andreas Nöth received his doctoral degree in Material Science and Engineering from Ecole Polytechnique Federal de Lausanne EPFL (Switzerland) in 2009. His thesis focused on the development of paraelectric ceramic thin films and microfabrication of devices based on these thin films. After his thesis, he joined Fraunhofer ISC in 2009. Since then, his focus has been on the development of Ceramic Matrix Composites. He is currently team manager of the section “Precursor Ceramics” that deals among other things with the development of fibre coatings and EBCs for CMC.

Alexander Konschak received his doctoral degree in experimental Petrology on CO_2 solubility mechanisms in silicate melts up to more than 1000 °C at University Bayreuth in 2009. Since 2008, he works at Fraunhofer ISC – Center for High Temperature Materials and Design on the research field of manufacturing and high temperature processing of CMC focusing on the Liquid Silicon Infiltration (LSI) of porous fibre preforms. In recent years, he is

focusing on the development of SiC/SiC using LSI and porous preforms. In detail, he investigated the microstructure and resistivity of coated SiC-fibres in silicon melts for example, using minicomposites.

Relevant infrastructure

Fraunhofer will contribute a significant infrastructure to the project to perform WP1 and WP6. Commercially available ceramic fibres will be used for the further processing to preforms within WP1. Due to the specific requirements in each task, preforms can be developed in different sizes and shapes according to the needs of the further CMC manufacturing, characterization and end-user.

The major items and technical equipment cover:

- Manufacturing devices for ceramic slurry recipes and characterization methods, e.g. rheometer
- Continuously working coating machine to apply oxide and non-oxide fibre coatings
- Filament winding machine with impregnation unit to produce fibre preforms, e.g tubes, based on endless ceramic fibres or UD tapes
- Continuously working roll-to-roll prepreg machine in order to produce 2D-fabric based prepregs
- Axial hot press to manufacture prepreg based CMC plates
- Coating device, e.g. spraying unit, for the application of EBC onto CMC substrates
- Several high temperature furnaces from 10 l up to 400 l for the pyrolysis and sintering of fibre preforms
- Non-destructive testing methods (X-ray, ultrasonic) for the quality control and inspection of fibre preforms during processing
- Characterization methods, e.g. SEM, LM, XRD for the quality control and improvement during processing

4.1.5. UOB

The University of Birmingham	
https://www.birmingham.ac.uk/schools/metallurgy-materials/index.aspx	
UK	*Higher Education*

Description

Founded in 1900, UoB was England's first civic university where students from all religions and backgrounds were accepted equally. With ~30,000 students from nearly 150 different countries, it was UK University of the Year in 2013-14 and ranks 122nd in the Shanghai Jiao Tong University World Rankings (2019-20) and 79= in the QS World University rankings (2019). The School of Metallurgy & Materials has 15 full Professors and 19 other academics. It has ~150 PhDs, 60 postdocs, ~200 undergraduates and 20 master's students. Research is divided across 4 themes, viz. Functional Materials Processing, Engineering Properties of Materials, Characterisation & Modelling and Alloy Processing; all but the last are relevant to this proposal. The School hosts the Interdisciplinary Research Centre in Materials Processing and the Centre for Electron Microscopy and has recently undergone a £35M (€49.5M) refurbishment programme. The aim of the ceramics research at UoB is to understand the processing / microstructure / property relationship to the point where materials with the desired properties can be designed in terms of the required microstructure and then manufactured. Work is underway on a wide range of ceramics and composites, including both electroceramics and structural ceramics and composites and both conventional and specialist processing routes are available for this work. UoB has expertise in both oxide and non-oxide ceramic materials and extremely good industrial links across a wide variety of end-applications. It has strong international collaborations within Europe and the USA in particular, including educational programmes. Outreach is extremely important as is technology transfer; the University has its own company, UoB Enterprise Ltd, to assist with the latter via licensing, spin-off creation and other routes.

UoB is internationally recognised for its expertise on the processing and characterisation of a wide range of ceramics and ceramic-based composites. A large number of papers have been written and very many talks, including Keynote, Plenary and invited, have been given at international conferences and workshops. UoB is currently involved with projects funded by both the UK Government and private industry in this field, and the PI has a track record in the field going back almost 3 decades.

Main tasks

In the CEM-WAVE project, UoB will be responsible for the processing and characterisation of the oxide-oxide composites. UoB will focus primarily on developing a new route to the manufacture of ox-ox CMCs, viz. the use of microwave chemical vapour infiltration to deposit oxide matrices within oxide powder impregnated oxide ceramic fibre preforms. They will also assist other partners with the characterisation of their materials.

Relevant publications/products/services/activities

Projects:

- AFRL $130,470 High temperature composites using microwave enhanced chemical vapour infiltration (2016-2019).
- ATI £269,042 Excite: Processing and manufacture of Ox-Ox CMCs (2018-2020). Total programme budget £9,344,041.
- DSTL MSA programme £97,976 Novel high temperature capable electromagnetic materials, with Rolls-Royce (2019-22).
- DSTL MSA programme £97,476 Production of ceramic materials for high temperature, low observable materials, with BAe Systems (2018-21).
- Rolls-Royce £137,074 CEMTEC(M): Production of ITAR-free Ox-Ox CMCs (2019-2020).

Publications:

- ‘SiCf/SiC fabrication by vacuum bagging, electrophoretic infiltration and microwave enhanced chemical vapour infiltration’, Jaglin D, Binner JGP, Prentice C, Shatwell R, Timms L and Westby W. Published on ECCM 9 conference proceedings CD, Institute of Materials, UK, (2000).
- ‘Reducing chemical vapour infiltration time for ceramic matrix composites’, Timms LA, Westby W, Prentice C, Jaglin D, Shatwell RA and Binner JGP. J Microscopy 201 [2] 316-323 (2001).
- ‘Densification mechanisms in SiCf/SiC composites by microwave enhanced chemical vapour infiltration’, Jaglin D., Vaidhyanathan B., Binner J, Prentice C and Shatwell B, Microwave and Radio Frequency Applications, Proceedings of the 3rd World Congress on Microwave and Radio Frequency Applications, pp 231-240, Folz DC, Booske JH, Clark DE and Gerling JF (Eds), American Ceramic Society, USA (2003).
- ‘Microwave heated chemical vapour infiltration of SiC powder impregnated SiC fibre preforms’, Binner JGP, Vaidhyanathan B and Jaglin D. Adv Appl Ceram. 112 [4] 235-241 (2013).
- ‘Creating Ultra-High Temperature Ceramic Matrix Composites By Microwave Or RF Assisted Chemical Vapour Infiltration’, 3rd Global Congress on Microwave Energy Applications (3GCMEA), Cartegena, Spain, July 2016.

***Products / Services / activities**:*

- UoB can offer materials characterisation, including SEM, TEM, FIB, EDX; XRD; µCT; and many other techniques, including thermoablation testing.

Key personnel

Prof Jon Binner, CEng FIMMM, FACerS, FECerS, FAmpere is Professor of Ceramic Science and Engineering at the University of Birmingham. The focus of his research is the generation of both the necessary scientific understanding and the required engineering solutions for the development of processing routes for advanced ceramic materials and composites that display technical and/or financial advantages over existing processes and which yield new or improved materials. It usually involves adopting a multidisciplinary approach. Over his career, which began in the early 1980s, he has attracted 136 research grants totalling ~£17.1M (€20.1M), supervised 35 PhD students and 2 MPhil students to successful completion, with 8 more PhD students currently being supervised and 3 being recruited, and also supervised 39 Postdoctoral Research Fellows with 5 currently ongoing and 2 being recruited. Since he recruited his first member of his research team, he is proud that 40% of them have been female and 60% have been BAME. His research has received both national and international recognition; he has given ~67 keynote, plenary and invited talks at international conferences (twice his paper has received a prize for being the best paper at a conference), whilst the Institute of Materials, Minerals and Mining (IOM3) has awarded him the Holliday Prize (1995), Ivor Jenkins Medal (2007), Verulam Medal & Prize (2011) and Pfiel Award (2017). In March 2016 a research programme that he worked on funded by MBDA, a major European defence manufacturer, was awarded an Innovation Award.

He has been the UK’s representative on the International Ceramics Federation since 2011 and the UK’s representative on the European Ceramic Society Council since 2010. For the latter, he was the leader of the R&D Working Group on its Permanent Executive Committee until July 2017 when he became the President-Elect. He took over as President in June 2019 (to June 2021, i.e. during Brexit). He was Vice-President, President and then Past President (the latter until Dec 2016) of the Institute of Materials, Minerals & Mining in the UK, each role lasting for 2 years. As President he chaired Council, whilst as Past-President he chaired the Management Board. During the last position, he oversaw the Institute moving from its home since 1972 to a new location across London. He was Vice Chair and then Chair (the latter until Oct 2016) of the UK Ceramic Society, each role lasting for 4 years consecutively. He is a member of both the Robert L. Coble Award for Young Scholars Award Committee and the newly formed Humanitarian Committee of the American Ceramic Society. He was a member of the Association for Microwave Power in Europe for Research and Education Committee (AMPERE) from 1999 – 2009, its secretary from 2002 – 2007 and during 2017 was honoured by being voted one of only five Fellows worldwide over the 26 years AMPERE has existed.

He has been on the Editorial Boards of 6 different scientific journals, has published ~230 research papers, edited or contributed to 19 books and holds 7 patents with an 8th filed. He has made 3 media appearances, two on BBC radio and one on BBC One. He has been a member of the EPSRC Peer Review College in the UK since 2000, an Honorary Professor at Kunming University of Science & Technology, China, since Sept 1999 and an Honorary Professor at Beijing University of Chemical Technology, China, since Nov 2006. He is the current Chair of the Engineering Strategic Advisory Team (EngSAT) for EPSRC and also sits on their Equality, Diversity & Inclusivity Strategic Advisory Group (EDI-SAG). He acts as a research proposal referee for many bodies, including the EPSRC, UK; National Science Foundation, USA; Australian Research Council; South African Foundation for Research

Development; Singaporean Science & Engineering Research Council; Deutsche Forschungsgemeinschaft; Science Foundation, Arizona, USA and the US-Israelli Science & Technology Foundation.

Relevant infrastructure

- A full range of ceramic green forming equipment
- High temperature air and controlled atmosphere furnaces
- Separate microwave-heated and RF-heated chemical vapour infiltration furnaces
- Hot isostatic press
- Rheological characterisation equipment for slurries
- Optical and a very wide range of electron microscopes, including SEM, FEGSEM, TEM, FEGTEM, STEM, etc, EDS and WDS, EELS, etc.
- XRD, including a hot stage, and X-ray micro computer tomography
- Purpose-built oxyacetylene and oxypropane torch testing facilities
- Grinding and polishing machines

4.1.6. Archer Technicoat Ltd

Archer Technicoat Ltd.	
www.cvd.co.uk	
UK	*SME*

Description

ATL is a privately owned, high tech company based in UK. It is specialised in the development of CVD processes and the construction of industrial CVD equipment. ATL develops, designs & builds all types of CVD reactor and specialises in high temperature, low pressure (HT-LP) reactors for production of ceramic coatings and ceramic matrix composite materials (CMCs). ATL has many years of experience in various forms of CVD and CVI, including experience of microwave heating and is a specialist in manufacture of bespoke CVD equipment for specific projects. In recent years it has produced several pieces of industrial scale CVI equipment for the EU, US and Asian market. ATL participates in EU funded R&D projects, with a long & successful track record stretching back over 25 years. We have worked with several of the project partners in previous projects.

Main tasks

ATL will contribute to WP3 activities on MW-CVI equipment. ATL will design, build, test and install CVI equipment components in collaboration with UNIPI, UoB. It will also provide experience and advice on CVI processing techniques, including microwave CVI. In fact, they will give a further implementation of these systems to bring a wide range of challenges. According to the system size, materials used and the required product, high-quality systems to produce CMCs will be guaranteed by ATL

Relevant publications/products/services/activities

Projects:

- FP7 project HELM. This project was led by the University of Pisa and in it ATL developed and built a unique pilot scale M-CVI equipment together with other partners. This equipment will continue to be used during the proposed project

Publications:

- Microwave Heated Chemical Vapor Infiltration: Densification Mechanism of SiCf/SiC Composites. September 2006, Journal of the American Ceramic Society 89(9):2710 – 2717
- High Temperature Properties of SiC and Diamond CVD-Monofilaments. July 2005, Journal of the European Ceramic Society 25(11):1929-1942

Products / Services / activities:

- Industrial supervision of PhD student at Loughborough University working on Microwave CVI. Calvin Prentice acted as this position
- Supply of microwave CVI equipment to a UK University. ATL built a lab scale MCVI equipment for the University of Loughborough

Key personnel

Calvin Prentice, Technical Director at ATL, has a degree in Natural Sciences (Chemistry) from Cambridge University and a MSc in Advanced Materials Technology from University of Surrey. He has worked in various fields of CVD since 1986 and leads the R&D team at ATL where CVI of CMCs is a regular product and under continuous development. He has been involved in previous microwave CVI projects with QinetiQ, Safran, University of Loughborough, University of Birmingham and the University of Pisa.

John Yeatman, Managing Director at ATL. He has a degree in Chemistry from Oxford University, has worked in the field of CVD since 1983 and specializes and in the production of equipment for ceramic matrix composites (CMCs). He leads the equipment production team at ATL.

Infrastructure

ATL have made many pieces of equipment capable of making ceramic composites. ATL have 6 CVD furnaces and various analytical equipment at their facility in the UK, as well as a dedicated team, space and workshop for assembling and testing new CVI equipment.

4.1.7. ENEA

AGENZIA NAZIONALE PER LE NUOVE TECNOLOGIE, L'ENERGIA E LO SVILUPPO ECONOMICO SOSTENIBILE – ENEA	
www.enea.it	
Italy	*RES*

Description

ENEA, the Italian National Agency for New Technologies, Energy and the Sustainable Economic Development, is a public undertaking operating in the fields of energy, environment and new technologies to support competitiveness and sustainable development. ENEA has approximately 2800 employees. In this project the department SSPT (Sustainability of territorial and productive systems) will be involved, with has approximately 600 employees, organized into several divisions. ENEA SSPT subteam has a unique background and over 30 years of experience in the development of circular economy, ceramic materials and recyclable materials. Specific expertise in this project is on composite materials development and characterization, accelerated ageing tests simulating expected working environment, prototype production, pilot lines engineering. Two laboratories are expected to be involved: SSPT-PROMAS-TEMAF (Faenza) for conventional CVI process, non-oxide CMC joining, mechanical and microstructural characterizations (before and after accelerating ageing).

Main tasks

The partner ENEA will coordinate WP4 devoted to mechanical and microstructural characterizations, with the facilities of TEMAF laboratory. TEMAF also possess a Chemical Vapour Infiltration plant, which might be involved in the preparation of performs of the development of joining.

Relevant publications/products/services/activities

Projects:

- MATISSE, - FP7-EURATOM, 2013-2017, focuses on materials' innovations for safe and sustainable nuclear. The project ended on 31st of October 2017. Claudio Mingazzini coordinated the WP3 (http://www.fp7-matisse.eu/wp3-p1-summary/) about advance fibre-reinforced composites for nuclear fission fuel cladding. The overall budget was of WP3 was about 1M€ and a TRL of 5 was achieved (ageing in expected working conditions, in IV gen Gas Fast Reactors).
- EEE-CFCC (2016-2018, POR-FESR Emilia Romagna Region, www.eee-cfcct.it): project coordinated by Claudio Mingazzini (ENEA TEMAF), about the exploitation of advanced fibre reinforced composites and preceramic prepreg development.
- FireMat (2019-2021, POR-FESR Emilia Romagna Region, www-firemat.it): project coordinated by Claudio Mingazzini (ENEA TEMAF), about the exploitation of preceramic prepreg for the development of fire resistant recyclable materials.
- C2CC Cradle-to-Cradle Composites (2019-2022, EIT RAW materials, upscaling project, www.c2cc-project.eu): project coordinated by Claudio Mingazzini (ENEA TEMAF), about the development of new semifinished materials for recyclable composites, the production of automotive prototypes (front bonnet of 500 Abarth model) and their mechanical assessment.
- Nextower (H2020, 2017-2020) project coordinated by ENEA, with the involvement of ENEA TEMAF about the development SiC Ceramic absorbers for high temperature (up to 900°C) Concentrated Solar Power.

Publications:

- Patent RM2014A000726(16 Dec 2014)., then PCT WO201698022 A1 and now being extended to EP "A method to manufacture an article made of a fibre-reinforced thermostructural composite", C. Mingazzini, A. Bedeschi (RiBa C. srl)" Authors: Claudio Mingazzini (ENEA), Andrea Bedeschi (RI-BA COMPOSITES - S.R.L.).

- "Optimization of a pyrolysis procedure for obtaining SiC-SiC f CMC by PIP for thermostructural applications" Claudio Mingazzini, Alida Brentari, Federica Burgio, Emiliano Burresi, Matteo Scafè, Luciano Pilloni, Daniele Caretti and Daniele Nanni, Advances in Science and Technology Vol. 77 (2013) pp 153-158.
- "Poly-Siloxane Impregnation and Pyrolysis of Basalt Fibers for the Cost-Effective Production of CFCCs" C. Mingazzini, M. Scafè, A. Brentari, E. Burresi, D. Caretti, D. Nanni, Advances in Science and Technology Vol. 89 (2014) pp 139-144.
- "Microstructural Characterization of SiC–SiC Joint by Raman Spectroscopy", G. Magnani A. Brillante L. Farina, L. Beaulardi, Journal of the American Ceramics Society Vol 87 (2004) pp 651-655 (joints based on patent RM2001A000707, Giuseppe Magnani et al, Procedimento per Giunzione di Materiali Ceramici, in particolare materiali ceramici a base SiC).
- Assessing Silicon Carbide Ceramics and Composites for Energy-Related Applications. DOI: 10.1109/EEEIC.2018.8494444, Conference: 2018 IEEE International Conference on Environment and Electrical Engineering and 2018 IEEE Industrial and Commercial Power Systems Europe (EEEIC / I&CPS Europe), C. Mingazzini, A. Rinaldi, M. Scafè, C. Sauder, J. Braun, M Steinbruck, Karsten Klemens Hansen.

***Products / Services / activities**:*
N/A

Key personnel
Claudio Mingazzini – (male, PhD in Chemistry, Master in Analytical Chemistry) is staff member at ENEA (Faenza), with over 25 years expertise on ceramic materials, the last ten of which fully devoted to the development of advanced long-fibre reinforced composites. In the last 5 years he has been coordinator of the following 5 projects/tasks, with an overall budget of over 5M€ (1) the coordinator of WP3 of FP7 project Matisse (2015-2017), devoted to SiC/SiC development and assessment for nuclear fission applications (fourth generation GFR reactors); (2) the coordinator of POR-FESR Emilia Romagnra Region EEE-CFCC (www.eee-cfcc.it, 2016-2018). (3) the coordinator of task 3.2 of H2020 project Nextower (2017-2020), about SiC application to high temperature (900°C). (4) the coordinator of POR-FESR Emilia Romagnra Region FIREMAT (www.firemat.it, 2019-2021). (5). the coordinator of EIT Raw Materials upscaling project C2CC, Cradle-to-Cradle Composites (www.c2cc-project.eu, 2019-2022). He will coordinate WP4 and ENEA TEMAF activities.

Giuseppe Magnani (male, Chemist) expert on silicon carbide based materials and Metallic Matrix Composites. He develops innovative methods to produce high performance components having high oxidation resistance up to high temperature. He authored 50 papers published in peer-reviewed journals and conference proceedings and five patents. He is the Coordinator of ENEA-SSPT-PROMAS_TEMAF lab.

Matteo Scafè (male, Aerospace Engineer) since 2007 at ENEA (Faenza) assist with pilot-plant engineering and coordinates activities about mechanical and thermomechanical characterization of composite materials for energy and aerospace applications, within National and European projects.

A shortlist of other collaborators in ENEA team will include (in random order): Alessandra Strafella (female – Engineer, expert of DOE approach), Selene Grilli (female – biologist, expert of TG-DTA), Antonino Coglitore (male – Engineer, reference person for creep characterisations), Elena Salernitano (female – Engineer, involved for SEM chacterisations), Federica Burgio (female – Engineer, expert of CVI processes and managing ENEA CVI plant), Francesca Mazzanti (female –Engineer, expert of oxide materials, alumina and zirconia based), Federica Bezzi (female – Chemist, expert of XRD characterizations of ceramic materials).

Relevant infrastructure
Characterization and qualification facilities (TEMAF):
- MTS machines for flexural and tensile tests, at room or higher temperature.
- Flexural creep (up to 1500°C) and RT intrusion tests facilities.
- Microstructural and physiochemical instrumentation (SEM-EDS; ICP-OES; TG-DTA; XRD).
- US non destructive tests.

Relevant pilot lines (TEMAF):

- CVD/CVI pilot line (internal volume diam. 300mm x height 700mm) for producing SiC and C coatings and fibre-reinforced ceramic composites.
- Filament winding facility, fully programmable on mandrels up to 300 mm of diameter and 1200 mm length.
- Warm pressing (up to 1300x2200mm) up to 200°C and Pyrolysis Pilot line (internal volume 1300x1200x300mm) working up to 900°C in controlled atmosphere, to produce fire resistant CMC-panels by PIP method.
- Furnaces of virtually any type and size for thermal treatments.
- 5 axes machining of molds up to the volume 1600x2200x600mm; 5 axes machining of ceramics on lower volumes.

4.1.8. CNRS

Centre National de la Recherche Scientifique (CNRS)	
www.cnrs.fr	
France	*Public Scientific and Technological Research Organization*

Description

The Centre National de la Recherche Scientifique (CNRS) (National Center for Scientific Research) is a public scientific and technological organization operating under the responsibility of the French Ministry of Higher Education and Research. Founded in 1939, CNRS is the largest fundamental research organization in Europe. CNRS has a budget of approximately €3.3 billion and employs some 31 570 staff, including 15,300 researchers and 16 270 engineers and technicians. It relies on more than 1 144 research units - mainly joint laboratories with universities and industry - located throughout France. 18 Regional Offices manage laboratories locally and liaise with local authorities and academic partners.

The Laboratory for ThermoStructural Composites (LCTS) is a joint unit of CNRS, University of Bordeaux, CEA and the Safran group. It has 31 years of experience in developing research on the most up-to-date high-temperature advanced ceramics and composites, both by experiments and models. Its main research topics are (i) Design and development of fibre-reinforced ceramic composites and of cellular ceramics, (ii) Their physicochemical and structural characterization at all scales, (iii) Testing their behaviour under mechanical, thermal and/or oxidative loadings, (iv) Theory and simulation of the structure and behaviour of these materials and of their fabrication processes. The present initiative has the agreement but no direct financial support from its industrial partners.

The Research Unit on Extreme Conditions & Materials: High Temperatures and Irradiation (CEMHTI) is an own research CNRS unit. It has 50 years of experience in developing high temperature facilities for multi-spectroscopic characterization of materials (Infrared and Raman spectroscopies, NMR…). Its main topics are (i) Multi-scale characterization of dynamical and structural properties of solids and melts up to extreme temperatures, (ii) Measurement of thermo-physical properties, (iii) Elaboration of glasses and glass ceramics, (iv) Numerical simulation of materials properties. In this laboratory, the “Optics&Thermics of Materials” Research Group has rare specific skills in measuring high-temperature emissivity of materials, including semi-transparent ones in the whole spectral range of thermal emission. It has also developed unique numerical tools to extract microscopic information from experimental data.

The Laboratory Interdisciplinaire Carnot de Bourgogne (ICB) jointly operated by CNRS, University of Burgundy and University of Technology Belfort-Montbéliard, counts 300 Physicists, Chemists, Engineers and Technicians in Dijon, Le Creusot, Châlon-sur-Saône & Belfort (Sevenans) campuses. Its main research topics are (i) Quantum systems and Photonics, (ii) Nanosciences and interfaces, (iii) Mechanical (iv) Numerical modelling of systems and materials, (v) Surfaces, Advanced materials, Durability. In the Process of Metallurgy, Durability, Materials (PMDM) team, thermal spray processes are developed and implemented for decades and have also been used as facilities to test materials under severe environment (high thermal load up to 90 MW/m^2, high temperature combustion, simulation of gas turbine environment with “in-flame generation” of ash particles).

Main tasks

CNRS-LCTS will be involved in several Tasks:

- Task 3.6 “Processing of ceramic inlays for increased thermal efficiency of CMC's samples”. LCTS will design ceramic inlays according to the necessary requirements, including heat transfer efficiency while retaining a moderate pressure drop of combustion gases, supervise their fabrication and ensure their delivery for joining inside the tubes.
- Task 5.1 “Micro-scale modelling of MW-CVI processes” and Task 5.3. Development of new Image-based numerical modelling using I.A. methodologies for the MW-CVI process” will be addressed by chaining together AI-based pattern recognition tools suited for the extraction of CMC fibrous reinforcements in X-ray tomographs to REV-scale modelling tools for (i) gas transfer by diffusion, Knudsen diffusion and viscous Darcian permeation, (ii) matrix deposition coupled to gas transfer, (iii) heat transfer by conduction and

radiation and finally (iv) complex permittivity. Global process optimization will be achieved through AI-based research of optimal processing conditions making use of the part-scale process simulation.

CNRS-CEMHTI will be involved in Task 4.3 “Measurement of Thermal Conductivity and Emissivity at High Temperatures” by performing normal spectral emissivity measurements on the materials investigated in the project. It will also contribute to the analysis of the spectra and it will advise on material properties improvement.

CNRS-ICB will be involved in Task 7-1 bis “Evaluation of CMCs samples in relevant environment of biogas” by designing and performing specifical tests on CMC samples under combustion environment representative of the working conditions of radiant tubes as defined by AM.

Relevant publications/products/services/activities

Projects:

- French Research Agency (ANR) & Aquitaine Region project ViSCAP (2017-2021, 595 k€) “Virtual Self-healing Composites for Aeronautic Propulsion”, PI : G. L. Vignoles
- French Aquitaine Region & Safran group project GaMA (2019-2022, 545 k€) “new Gas routes for Aeronautic MAterials preparation”, PI: S. Jacques, LCTS, WP leader: G. L. Vignoles

Publications:

- M. Pelanconi, M. Barbato, S. Zavattoni, G. Bianchi, G. L. Vignoles, A. Ortona, “Thermal design, optimization and additive manufacturing of ceramic regular structures to maximize the radiative heat transfer”, Materials & Design (2019) vol. 163, 107539
- J. Bénézech, G. Couégnat, “Variational segmentation of textile composite preforms from X-ray computed tomography”, Composite Structures 230, 11496 (2019)
- G. L. Vignoles, “Modeling of Chemical Vapor Infiltration Processes”, Chapter 17 of “Advances in Composites Manufacturing and Process Design “, P. Boisse Ed., Elsevier Woodhead Scientific. ISBN-13: 978-1-78242-307-2, pp. 415-458. (2016)
- O. Rozenbaum, C. Blanchard, D. De Sousa Meneses, "Determination of high-temperature radiative properties of porous silica by combined image analysis, infrared spectroscopy and numerical simulation", International Journal of Thermal Sciences 137 552-559 (2019)
- I. G. de Arrieta, T. Echaniz, R. Fuente, L. del Campo, D. D. Meneses, G. A. Lopez, M. J. Tello, "Mid-infrared optical properties of pyrolytic boron nitride in the 390-1050 °C temperature range using spectral emissivity measurements", Journal of Quantitative Spectroscopy and Radiative Transfer 194 1-6 (2017)
- M. Molière, C. Verdy, US patent N°US 2013/0306154 A1, nov. 21. 2013: “Process to create a collision between a stream of gas and particles and a target”,
- C. Liégaut, P. Bertrand, L. Maillé, F. Rebillat, “UHTC bases matrix as a protection for C/C composites for very high temperature use: original manufacturing and oxidation resistance” 15th Conference & Exhibition of the European Ceramic Society Budapest, Hungry –EcerS2017
- C. Verdy, C. Coddet, G. Montavon, M. Molière, M. Vierling, D. Sokolov, ‘Physics and Chemistry of Ash in Gas Turbines: The HVOF Technique as a Powerful Simulation Tool’, in Proceedings of ASME Turbo Expo 2012, ASME Proceedings (presented at the GT 2012, Copenhagen, Denmark: ASME, 2012), pp. 529–534

***Products / Services / activities**:*

- Gentools software suite: Virtual material strategy applied to CMCs [Composite Structures 230, 11496 (2019)].
- CVI modelling software suite: DMC (Densification Monte-Carlo) [Comput. Mater. Sci. 50(3), 1157-1168 (2011)] and LIRWA (Large-scale Infiltration by Random Walks) [Comput. Mater. Sci. 50(6), 1871-1878 (2011)]
- Thermal modelling software: SHTRAC (Simulation of Heat Transfer by Radiation And Conduction) [Int. J. of Heat & Mass Transfer 93, 707–719 (2016)]

Key personnel

Pr. Gerard L. Vignoles will be the WP5 leader. University of Bordeaux, head of LCTS. Graduated from Ecole Normale Supérieure (rue d'Ulm, Paris), he joined LCTS at its beginning as a PhD student. After his defence in 1993, awarded by several French prizes, he has been hired in 1994 as Assistant Professor of the Bordeaux University; in 2003 he became full Professor; in 2016 he was appointed head of LCTS. During 30 years he has developed and maintained there an activity covering image-based physicochemical modelling of the fabrication and behaviour in use of thermo-structural composite materials, supervising and co-supervising many industry-funded PhDs (29 + 4 under way). He has been the PI of the French Research Agency (ANR) project PyroMaN (2010-2015, 470 k€) and is currently the PI of the ANR+Aquitaine Region project ViSCAP (2017-2021, 595 k€). He has developed several pieces of software for the modeling of Chemical Vapor Infiltration at various scales, using various numerical methods and also has recently developed a new numerical method for the image-based 3D modelling of coupled radiative-conductive heat transfer in porous media and applied it to ceramic foams. Editor of the Journal of the European Ceramic Society, he is the author of 141 papers and has a h-index of 26 (according to Scopus).

Dr. Bruno Dubroca, CEA Research Director at LCTS since 2018, is a specialist of mathematical modeling and numerical simulation techniques, with special focus on propagation equations solving by varieties of methods. He is the author of 65 papers and has a h-index of 15 (according to Scopus).

Pr. Francis Rebillat of University of Bordeaux, is a specialist of oxidation & corrosion of ceramics, including self-healing Ceramic-Matrix Composites, and of ultra-refractory thermal protection coatings, managing both experimental and modelling aspects, supervising many industry-funded PhDs (21+2 under way). He is the author of 70 papers and 6 patents and has a h-index of 167 (according to Scopus).

Dr. Leire del Campo, Associate professor at the University of Orleans and researcher at CEMHTI since 2011, is a specialist in infrared and impedance spectroscopies and works on the development of experimental devices for the characterization of materials. She is the author of 40 papers and has an h-index of 14 (according to Scopus).

Dr. Pierre Bertrand, Research Engineer at the University of Technology Belfort-Montbéliard since 1998 has developed skills and know how in coatings and surface treatments. Its main activities are based on industrial partnerships and concern thermal barrier/EBC coatings for aeronautical and aerospace industries, wear and abrasion resistant coatings for food industry, protection against corrosion for nuclear industry. He is the author of 14 papers and has a h-index of 8 (according to Scopus).

Dr. Christophe Verdy, Research Engineer at the University of Technology Belfort-Montbéliard since 1998 has developed skills and know how in thermal spray coatings and thermal tests. Its main activities are based on metallic additive manufacturing by the cold spray process and thermal-erosive tests under combustion environment. He is the author of 14 papers and has a h-index of 12 (according to Scopus).

Relevant infrastructure

- LCTS has a local computational cluster containing 84 core CPUs, allowing simulations on numerical domains containing millions of DOFs; if larger computations are required, access to the Aquitaine Regional HPC facility (MCIA) and to the national HPC facility of CNRS (GENCI) will be made.
- LCTS has access to the local CNRS Service Unit for Materials Characterization (PlaCaMat) for the acquisition of the necessary X-ray Computerized Micro-Tomography data.
- LCTS possesses several high-temperature testing facilities, including a universal mechanical testing machine equipped with a heating stage and a supply of $O_2/H_2O/N_2/Ar$ gas mixtures and a high-pressure vessel able to deliver temperatures up to 1400°C, total pressure up to 30 bar and up to 1 bar water vapor pressure.
- CEMHTI has developed a unique emissivity measurement setup for characterizing the thermal radiative response of materials form far-infrared to visible range and from room temperature up to 3000K.
- ICB has developed specific systems based on thermal spray devices for evaluating the materials response under very severe environment (high thermal fluxes up to 90 MW/m^2 with a plasma jet and up to 10 MW/m^2 with a combustion flame and a specific chemistry).

4.1.9. POLITO

Politecnico di Torino	
www.polito.it	
Italy	*University*

Description

POLITECNICO DI TORINO was founded about 150 years ago and represents a leading public university, in Italy and Europe, in technical-scientific teaching, research and technological transfer with35000 students from more than 100 countries. Politecnico di Torino is part of some of the major European interuniversity networks, such as CESAER, CLUSTER, E.C.I.U, EUA, and CMU. These contacts will permit to spread and disseminate information and project results to a wider number of institutions. With more than 210 European and International projects (64M€ since 2014) and about 167 Horizon 2020 projects (60M€), POLITO can count on a great experience both in scientific projects and administrative management. POLITO is involved as coordinator and partner in several EU projects promoting the development of innovative doctoral programmes involving both academia and enterprise. In November 2013 the European Commission awarded the 'HR Excellence in Research' logo to Politecnico di Torino, remarking the strong commitment in the implementation of the principles of the Charter and Code aimed at creating a high quality academic environment where researchers and students from all over the world can study, work and collaborate on innovative and challenging projects.

Main tasks

POLITO will participate to this projects by leading WP6 (Joining and coating of the produced CMCs samples); POLITO will also contribute to the set-up preform materials for MW-CVI processes (WP1), oOptimization of the MW-CVI process for non-oxide and oxide CMCs (WP3) characterization of joined components (WP4), testing and validation of CMCs prototypes (WP7), LCA and risk assessment (WP8) and dissemination, communication and exploitation (WP9).

Previous experience relevant to those tasks.

POLITO participated as coordinator or partner to seven EU projects regarding joining and mechanical testing. POLITO has been involved for many years in the study, manufacturing and mechanical testing of joints for ceramic matrix composites in collaborations with several national and international research laboratories; the results of these collaborations are reported on several scientific papers and international conference presentations listed at *www.composites.polito.it*.

POLITO was Project Coordinator or Principal Investigator of 27 National and EU projects on joining and coatings of advanced materials, from 1992-present: among them, 17 are EU projects, 8 of them she acted as Project Coordinator. Considering the fierce competition (less than 1.5% success in some EU calls), this results is to be considered of high relevance; the approximated amount of funding is of the order of magnitude of several million euro. Some relevant (Project coordinator or Work Package coordinator) ones are cited below:

- Human Capital and Mobility (1992-1994): ERBCHBGCT920035) (1996-2000)
- GLAMOROUS Glass-based modulators, routers and switches IST-2000-28366 (coordinator work package on glasses) (2000-2003)
- ODUPE Optical Devices Using Photosensitivity for their Elaboration IHP-RTN-99-1 (partner)(2000-2003)
- GLAPHOVIS Glasses for Photosensitivity in the Visible IST-2001-37069 (coordinator) (2002)
- EU-ASIALINK CN04-81260 (coordinator) (2004-2007)
- KMM-NoE (502243-2) (coordinator work package on joining) (2004-2008)
- EXTREMAT-IP (partner) (2005)
- FEMAS (partner) (2008-2011)
- MATRANS (partner) (2009-2013)
- GLACERCO (ITN 264526, coordinator) (2011-2015)
- NASLA (FP7-SME-2010-1, 262209,coordinator) (2010-2012)
- INNVIN (FP7.NMP partner – 2012 – 2015):
- JOLIE (FP6 MATERA, coordinator) (2012-2015)

- ADMACOM (FP7-FoF, coordinator) (2013-2016)
- MATISSE (FP7-Fission, partner) (2013-2017)
- COACH (H2020-partner)(2015-2018)
- SMS (H2020-partner) (2015-2017)
- Il Trovatore (Euratom-partner)(2017-2021)
- NEXTOWER (H2020-partner)(2016-2019)
- GrInHy (H2020-partner) (2016-2019)

Relevant publications/products/services/activities

Projects:

- European project **ADMACOM** FP7 –FoF.NMP.2013 GA 609188 (*Advanced manufacturing routes for metal/Composite components for Aerospace*). Coordinated by POLITO. The aim of ADMACOM was to develop innovative manufacturing technologies based on advanced design of interfaces and of joining materials for aerospace components
- European Project NMBP-17-2016 **NEXTOWER** (*Advanced materials solutions for next generation high efficiency concentrated solar power (CSP) tower systems for next generation high efficiency concentrated solar power (CSP) tower systems*). The main contribution is in providing joining solutions for ceramic solar receivers and in their characterization.
- European Project H2020-MSCA-ITN-2014 - GA 642557 **CoACH** (*Advanced glasses, Composites And Ceramics for High growth Industries*). POLITO coordinated this European Training Networks project. One of the topics was the joining of similar/dissimilar materials.
- European Project MATERA -ERANET (Ministry of Education, Universities and Research –MIUR) 2227 **JOLIE** (*JOining of LIghtweight alloys to advanced FGM mEtal-ceramic materials*). Research activity focused on the study of joining materials and processes for Al-alloy /ceramic brake systems.
- **SMS** - Sandwich material and structure (H2020) 2016-2018. POLITO was in charge of the development of joining material and processing for the aerospace industry.

Publications:

- Joining and mechanical testing of oxide/oxide (Nextel ™610/alumina-zirconia) ceramic composites AKRAM M. Y., CASALEGNO V, FERRARIS M PUCHAS G., KRENKEL W, ROSZEITIS S (2019) JOURNAL OF THE EUROPEAN CERAMIC SOCIETY, 39, Issue 7, pp 2510-2517
  DOI: 10.1016/j.jeurceramsoc.2019.02.020.
- Joining and testing of alumina fibre reinforced YAG-ZrO2 matrix composites, AKRAM M. Y., FERRARIS M., CASALEGNO V., SALVO M, PUCHAS G. , KNOHL S., KRENKEL W. (2018), JOURNAL OF THE EUROPEAN CERAMIC SOCIETY , 38, pp.1802- 1811
  DOI: 10.1016/j.jeurceramsoc.2017.11.026
- Pressure-less joining of C/SiC and SiC/SiC by a MoSi2/Si composite GIANCHANDANI, P.K., CASALEGNO V., SMEACETTO, F., FERRARIS, M. (2017) INTERNATIONAL JOURNAL OF APPLIED CERAMIC TECHNOLOGY, 14 (3), pp. 305-312.
  DOI: 10.1111/ijac.12631
- Joining of C/SiC composites by spark plasma sintering technique, RIZZO S., GRASSO S., SALVO M., CASALEGNO V., REECE M.J., FERRARIS M.; (2014), JOURNAL OF THE EUROPEAN CERAMIC SOCIETY, Volume 34, Issue 4, pp 903–913
  DOI: 10.1016/j.jeurceramsoc.2013.10.028
- Characterisation of joined surface modified SiCf/SiC composites, CASALEGNO V., VALENZA F., BALAGNA C., SEDLAK R., GIRMAN V. , SALVO M., DE LA PIERRE DES AMBROIS S., FERRARIS M.; (In: CERAMICS INTERNATIONAL. - ISSN 0272-8842. - (2019), in press.

Products / Services / activities*:*

- Ferraris, M., Casalegno, V., Salvo, M. Process to join carbon based materials to metals and its applications, (2005) Patent n. EP2214732. Publication info: EP1685079 (A1) — 2006-08-02. International Publication number WO 2005/037734 A1. Politecnico di Torino

Key personnel

Valentina Casalegno is Associate professor of Materials Science and Technology at Politecnico di Torino (Italy). After completing her PhD in Materials Science and Technology in 2006, her research activity has been mostly dedicated to design, fabrication and characterization of joining materials and joints for advanced ceramic composites and ceramic/metal components. She is also working on developing, characterization and testing of materials ("custom" glasses, glass-ceramics and their composites, in the form of bulk-, joining-, coating-, porous- , thin films- and fiber-materials) for extreme applications and energy conversion (aerospace, automotive and nuclear application). She is part of a team that aims at fostering high level research in Materials Science and Engineering through the participation to European projects and the cooperation with the most qualified international research centers and companies.

Monica Ferraris is Full Professor of Materials Science and Technology at the Politecnico di Torino, Torino, Italy. She received a M.S. in Solid State Chemistry in 1985 from the University of Torino, Italy. In 1985, M. Ferraris joined the Italian Telecom Research Centre (CSELT-Torino). In 1990 she moved to Fiat Research Centre (CRF-Torino) where she worked on metal and ceramic matrix composites for automotive applications. She joined Politecnico di Torino in 1991, where she is now Professor of Materials Science and Technology, group leader (http://www.composites.polito.it) and Head of the Institute of Materials Engineering and Physics at DISAT (Department of Applied Science and Technology). Her academic and research interests include joining of glasses, ceramics and composites for energy production, photonics, biomedical applications and waste management. She has more than 220 peer-reviewed journal publications and patents in these areas. M. Ferraris is member of the Engineering Ceramics Division of The American Ceramic Society since 1995.

Milena Salvo: is Full Professor of Advanced Ceramics and Materials Science and Technology. She has considerable experience of international research on advanced materials. Her expertise is in advanced ceramics and composites for energy production, joining and coating for high-temperature applications and waste management. She is member of The American Ceramic Society and is co-author of more than 160 papers in international journals, conference proceedings, book chapters and 4 patents (h-index: 34).

Relevant infrastructure

The DISAT is well furnished with a series of equipment that might be useful for the production and characterization of composites, nanocomposites, fibres, and for surface modification, joining, characterization and mechanical testing; the facilities are available for the project and can be seen at http://www.composites.polito.it/ . The research group is part of J-TECH@POLITO, Advanced Joining Technologies at Politecnico di Torino, an inter-departmental, multi-disciplinary research center based at POLITO involving three departments and 25+ staff members and related facilities for joining and testing, http://www.j-tech.polito.it/facilities .

4.1.10. IRIS

IRIS Technology solutions, SL.	
http://www.iristechnologygroup.com/	
Spain	*SME*

Description

IRIS Technology Group (http://www.iristechnologygroup.com/) is an advanced engineering Group that was established in Barcelona in 2007. Since 2011, it also operates from Dublin (Ireland) whereas the Barcelona office is referred to as IRIS Technology Solutions S.L. It specialises in the manufacture and integration of Process Analytical Technology (PAT)-based real-time quality monitoring solutions for the process industries and IoT systems for smart manufacturing, predominantly targeted at the food, pharmaceutical and chemical industries. In particular, our product line comprises proprietary state-of-the-art IR analysers for at-line and in-line process monitoring.

On the one hand, our solutions are based on the latest developments in photonics in the fields of MEMs, Near Infrared Spectroscopy, Hyperspectral Imaging, and Raman. On the other hand, we leverage ICTs, whereby we apply Machine learning, Artificial Intelligence and Data Mining techniques for predictive and active control, as well as for decision support systems. In addition to photonics-based analysers, we also work with IoT sensors for enabling equipment and machinery for smart manufacturing and transitioning industry to the Industry4.0 paradigm. Our solutions are cloud-based for mobility, remote monitoring and multi-site applications.

Our Innovation Division which is governed by an Innovation Committee comprising Senior Management, and an Innovation Unit for managing and executing the companies' innovation roadmap, has 3 core strategic lines, for shaping the Factory of the Future:

1. New optical photonics-based technologies and PAT systems for production process optimisation
2. ICTs (Artificial intelligence / machine learning, IOT & cloud platforms)
3. Circular and Bio-based Economy: digitalization and waste valorisation

In addition to running an internal R&D programme, as well as carrying our various joint private projects with industry partners, our Innovation Division is highly experienced in working in collaborative R&D projects in H2020 (both as coordinators and partners), principally in the SPIRE, FoF, and BBI programmes. The company was established in 2007 and currently employs 60+ staff with a turnover of € 5 M. It is equipped with engineering workshops (3), food testing and wet chemistry laboratory, optics labs (2), and an electronics/telecommunications lab. IRIS is ISO 9001:2008 certified and also is certified under UNE-EN-ISO 166002: Management Systems R&D&i. We are also certified by the local (Catalonian) government as a provider of technological R&D services (Tecnio Certification). **IRIS is a member of the following sectorial networks: A.SPIRE (honorific member of the board),** SPIE and SECPHO, in addition to regional industrial collectives such as Smart Space (association of companies and organisations that specialise in the 'smartification' of industry).

Main tasks

IRIS will select sensors with potential for in line implementation to support the optimal CMC process and later use phase of the CMC in WP3 and 7 respectively. They will also contribute to the dissemination and exploitation planning of the results.

IRIS has a matrix of multidisciplinary knowledge groups in the company strong expertise in data science, biopolymers, chemiometry, spectroscopy among many others. With its team and resources, including electronics engineers, mechatronics, mechanical, industrial design engineers, system engineering, application programming and web development, chemists, biologists, etc. the group can carry out full system design and engineering, including electronic design and assembly, firmware and software development, electrical harnesses and interconnections, mechanical design for enclosure and prototype integration, as well as installation, validation and industrialisation of prototypes and equipment in production plants.

Relevant publications/products/services/activities

Publications:

- E. Bugnicourt, T. Kehoe , M. Latorre, C. Serrano, S. Philippe and M. Schmid, Recent Prospects in the Inline Monitoring of Nanocomposites and Nanocoatings by Optical Technologies, Nanomaterials 2016, Review, 6(8), 150; doi:10.3390/nano6080150
- "MONITORING INNOVATION: Optical systems to support the production of surfaces with advanced functionalities". Elodie Bugnicourt, Severine Philippe, Tim Kehoe, Laura Rodriguez, Simona Neri, European Coatings Journal, 06 – 2019, pp 42-48.
- A. Rosales, O. Mc Nerney. Sensing Technology. Chapter 6. Resource Efficiency of Processing Plants. Monitoring and Improvement. Edited by Stefan Krämer and Sebastian Engell. Wiley-Vch. Print ISBN: 978-3-527-34074-3
- Conference ICIEMPM 2016: Towards Automatic Calibration of In-Line Machine Processes, 18th International Conference on Industrial Engineering and Manufacturing Process Modeling, Nettleton, D.F., Bugnicourt, E., Wasiak, C., Rosales, A., London, United Kingdom, July 28-29, 2016.
- "Injection moulding process optimization by online material monitoring". E. Bugnicourt et al., October 2012. www.packagingconnections.com

Projects:

- OPENMIND, H2020 FoF, GA 680820, On-demand production of entirely customised minimally invasive medical devices
- SUPREME, H2020 SPIRE, GA 768612, Sustainable and flexible powder metallurgy processes optimization by a holistic reduction of raw material resources and energy consumption."
- PROPAT, H2020 SPIRE, GA 637232, Robust and affordable process control technologies for improving standards and optimising industrial operations
- LIGHTME, H2020 NMBP, GA 814552, An Open Innovation Ecosystem for upscaling production processes of lightweight metal alloys composites
- MULTICYCLE, H2020 SPIRE, GA 820695, Advanced and sustainable recycling processes and value chains for plastic-based multi-materials

***Products / Services / activities**:*
N/A

Key personnel

Dr. Ing. ELODIE BUGNICOURT. *(female)* Masters in Material Science & Engineering (2002), PhD in Polymers and Composite Materials (2005), both from National Institute of Applied Science (INSA), Lyon, France. Leader of the Eco-Materials business unit in IRIS since 2009. Coordinated several European projects in the fields of bio-based packaging materials development, process monitoring, surface treatment. Previous industry experience: R&D in chemical industry especially related to reactive polymer formulations: i) Arkema (ex. Cray Valley, 2002, Spain) in the field of polyester powder coating; Wacker Chemie (2002-2005, Germany and France), Silicone division that funded her PhD thesis: development of nanocomposites based on an epoxy matrix and silica; iii) Ashland (2006-2009, located sequentially in US, Spain and France) in the Composite Polymers division, focussing on bio-sourced resins, flame retardant resins, SMC resins, PU and epoxy structural adhesives. Led and participated in many R&D projects aimed at developing new formulations, understanding mechanisms and structure-properties relationships, developing new analytical methods and modelling (for improved quality management); as well as collaborations with academic labs and other players in the supply and value chain. Author of several papers in peer-reviewed and technical journals, and has presented her work in numerous international conferences.

Dr. TIMOTHY KEHOE (Male): VISUMPalm Product Manager, as well as the Technical Lead at Dublin Facilities. PhD in Engineering, University College Cork, Ireland- developed an IR photometric measurement system for thermal diffusivity in anisotropic materials. Masters in Optoelectronics, Queens University Belfast.

Dr. CUAUHTEMOC ARAUJO (Male) Chemometrician. Degree on Physics, master degree in Material Sciences, and doctorate degree in Applied Sciences. Conducted research in the fields of materials sciences, optical spectroscopies and chemometrics. More than 20 years of experience working on a wide range of biological and non-biological applications of vibrational spectroscopies (i.e. Raman, MIR and NIR). Also worked closely with other characterization techniques such as: UV/Vis and fluorescence spectroscopies.

DR. LAURENT PHILIPPET (Male) Scientist. Bachelor's degree in science, speciality applied physics (Université Paris Sud). Engineer's degree in Optical Science (Institut d'Optique Graduate School). PhD in Optical Science (University Paris XIII). Experimental physicist specialised in optics and photonics with specific skills in laser physics, spectroscopy techniques, microscopy, interferometry and optical design.

Dr. DAVID NETTLETON. (male) Data Mining Analyst. Has a Ph.D in Artificial Intelligence from the Universitat Politécnica de Catalunya, Barcelona, Spain, M.Sc. In Computer Software and Systems Design from the University of Newcastle upon Tyne, U.K., and B.Sc. (Hons) Computer Science from the University of Wolverhampton, U.K. Has over 20 years of experience of applying data mining algorithms (including neural networks, rule induction) and statistical techniques (including ANOVA, regression, principal components) in a diversity of data analysis and modeling projects, such as medical diagnosis/prognosis, fraud detection, customer loyalty and segmentation, online user web search behaviour and data privacy. As well as commercial projects, he has conducted research and lectured at two universities. He has published a diversity of articles in specialised data analysis journals/congresses and three books. He has participated in several competitive multi-partner research projects and has a European patent application for a graph matching algorithm.

Dr. JOSE ANTONIO IBARRA FUSTE (male). Scrum Master. PhD. in Signal Theory and Communications, Universitat Politècnica de Catalunya, Spain (2014), Thesis title: "Design Rules and Optimization of Electro-Optics Modulators Based on Coplanar Waveguides"; the project dealt with the development of new models to design high speed electro-optical modulators based in LiNbO3, where a novel software to design and optimize these modulators was created. Master Degree in Telecommunication Systems from Universidad de Oriente, Cuba (2005). Telecommunication Engineer Degree from Universidad de Oriente, Cuba (2003). Author of several papers in journals of high international impact. Experienced in optical communication softwares as Virtual Photonic (VPI), OptSim, and in electromagnetic structures simulators as High Frequency Structural Simulator (HFSS) and ADS. Experience for more than 5 years in many programming languages.

Relevant infrastructure

Spectroscopic Laboratory equipped with several instruments, namely, a FTIR-ATR (Interspectrum), 2 UV-VIS (PerkinElmer and Ocean Optics), an Extended-SWIR (StellarNet) and a NIR (Mightex) spectrophotometers as well as a SWIR Hyperspectral imager (Specim-Xenics). The associated staff are properly trained and skilled in chemometrics modelling by using a commercial suite (SOLO+MIA from Eigenvector) as well as ad hoc non-commercial chemometric tools (Matlab and Python toolboxes).

Electronics, mechanical and optical workshops and laboratories equipped with up-to-date hardware and tools: signal generators, wide band oscilloscopes, PCB design [Altium] and building tools, development kits for programming microcontrollers, CNC-milling machines for electronic and mechanical prototyping, and CAD software [SolidWorks], among others.

IRIS' optical engineers work with ZEEMAX software for optimally designing of optical systems and to setup the designs by means of advanced optical components and facilities (optical tables, windows, lenses, lasers and light sources and related "opticalware").

Food and wet chemistry lab equipped with a DSC, a Surface Energy analyser, an HPLC, a Cobb tester, microbiological equipments, microspcope and microtome for structural analysis, migration cells, colourimeter (for colour and opacity measurement), Ford cup viscometers, water vapour permeation cells, turbidimeter, climatic chambers, manual and automatic coaters, electrospray, different drying technologies, plasma surface activation, etc.

4.1.11. AM

ArcelorMittal Innovación, Investigación e Inversión, S.L.	
Spain	*AMIII*

Description

ArcelorMittal Innovación Investigación e Inversion S.L. (AMIII) is a company that belongs to ArcelorMittal Group, the world's leading steel company, producing 10% of the world's crude steel output, and the leading supplier of quality steel products in all major markets including transport, construction, household appliances and packaging. ArcelorMittal is present in more than 60 countries, 32 out of them are actually European countries.

AMIII is located in ArcelorMittal R&D Centre in Asturias, which is an active member at different research networks. Research includes advanced process control and monitoring systems at industrial plants, energy, environment and recycling research, measurement technology development, intelligent systems and novel data mining techniques. AMIII is specialized in the implementation of new technologies in different phases of the steelmaking process, becoming a reference for the development of innovative technology projects at the regional, national and European level, due to its close collaboration in the development of projects with ArcelorMittal España and other European steel companies of the Group ArcelorMittal. In addition, there is a close contact between AMIII and different plants of the company in order to guarantee that solutions developed can be easily installed, tested, and finally disseminated in all the group installations that can take advantage of them. Among other facilities, the R&D Centre has several pilot plants for product processes simulation, mechatronics, a vision laboratory, etc. AMIII has also participated in a wide variety of long-term regional, national and European projects.

ArcelorMittal has an active presence in the main European and National technology platforms, committees, clusters and other initiatives related with steel: PTFE (Spanish Railway Technology Platform), PLATEA (Spanish Steel Technology Platform), ESTEP (European Steel Technology Platform), Worldsteel (World Steel Association), SPIRE (Sustainable Process Industry through Resource and Energy Efficiency), etc.

Main tasks

ArcelorMittal will define the technical and specific requirements for CMCs. ArcelorMittal will contribute by giving technical information about the state-of-the-art of heating elements and the currently used materials. ArcelorMittal will perform the validation of the CMCs samples for future and fluctuating extreme conditions. These tests will be done in the radiant tubes pilot plant. A special support must be designed and constructed to carry out the tests of the samples, because the foreseen diameter and length of the samples are smaller than the radiant tubes' ones. Once the sample is ready, a test campaign will be carried out in which the working conditions will be varied. Different furnace start-stop cycles will be carried out, combining different heating ramps and target temperatures. The composition of the gas used as fuel will also be varied to evaluate possible effects on the material in case of using alternative fuels to the natural gas.

Relevant publications/products/services/activities

Projects:

ArcelorMittal has an active presence in the main European and National technology platforms, committees, clusters and other initiatives related with steel: PTFE (Spanish Railway Technology Platform), PLATEA (Spanish Steel Technology Platform), ESTEP (European Steel Technology Platform), Worldsteel (World Steel Association), SPIRE (Sustainable Process Industry through Resource and Energy Efficiency), etc.

Some projects that were approved by the European Commission are:

- WAVISURF: Online Industrial waviness measurement (RFSP-CT-2011-00013)
- INTELLUB: Increased mill capability and performances by the application of intelligent lubrication (RFSR-CT-2013-00008)

- INTEGRAL: INitiative to bring the 2nd generation of ThermoElectric Generators into industrial ReALity (720878)
- OPTIPER: Optimisation of permeability bars to customise sinter plants on changing demands (RFSR-CT-2012-00004)
- HELNOX-BFG : High Efficiency Low NOx BFG Based Combustion Systems in Steel Reheating Furnaces (RFSR-CT-2012-00010)
- PREVENTSECDUST: Prevention of secondary dust emissions in ironmaking plants using dust suppressants" (RFSR-CT-2015-00027)
- I-ThERM: Industrial Thermal Energy Recovery Conversion and Management (680599)
- PROTEUS: Scalable online machine learning for predictive analytics and real-time interactive visualization (687691)
- CYBER-POS: Virtual Design of Cyber-Physical Production Optimization Systems for Long Production Factories (709669)
- SPOTVIEW: Sustainable Processes and Optimized Technologies for Industrially Efficient Water Usage (723577)
- INSPIREWATER: Innovative Solutions in the Process Industry for next generation Resource Efficient Water management (723702)
- WHAM: Water and related energy Hub Advanced Management system in steelworks (800654)
- GRAPHENE: Graphene Driven Revolutions in ICT and Beyond (649953)
- BAMBOO: Boosting new Approaches for flexibility Management By Optimizing process Off-gas and waste use (SEP-210505375)
- INTEGRADDE: Intelligent data-driven pipeline for the manufacturing of certified metal parts through Direct Energy Deposition processes (820776)
- BURNER 4.0: Development of a new burner concept: Industry 4.0 technologies applied to the best available combustion system for the Steel Industry
- SAFEDEWPOINT: Acid dew point and corrosion sensors for dynamic waste heat recovery from steel mill flue gases
- ONTOTRANS: Ontology driven Open Translation Environment
- C4U: Advanced Carbon Capture for steel industries integrated in CCUS Clusters

Publications:

- On-line industrial waviness measurement for process and quality control - ESTAD 2014, April 7 & 8 - G. Moréas (CRM Group, Liège), Belgium A. Garcia Martino (ArcelorMittal R&D Asturias), Spain, E. Montagna (Tata Steel Europe SEGAL), Belgium, W. Bilstein (Amepa Aachen), Germany

***Products / Services / activities**:*

<u>Key personnel</u>

D. Ángel García Martino is working in ArcelorMittal (R&D Department) since 2008. He was previously working as associate professor at University of Oviedo (Spain), working in the group of Signal Theory and Communications (Department of Electrical Engineering (DIEECS)). He has worked in several R&D projects developing and improving new processes in the plant and has a big experience in modelling processes to analyze the future impact of implementing a new solution in the plant. He has developed several R&D projects oriented to improve the galvanizing process: heating and cooling models, improvement of lifetime of key parts, development of new measurement systems, etc. He has participated in several continuous improvement workgroups in collaboration with the site technical team.

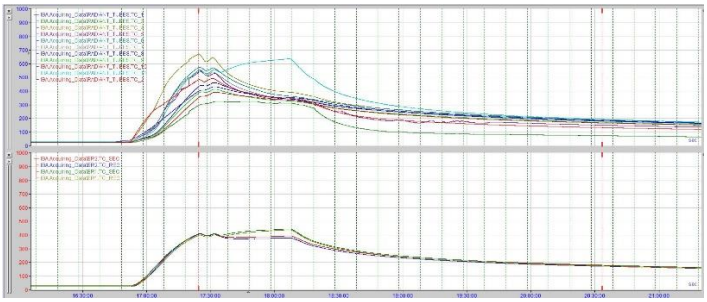
He was member of the Scientific Committee in International and National Surface and Heat Treatment Congress. Some papers published:

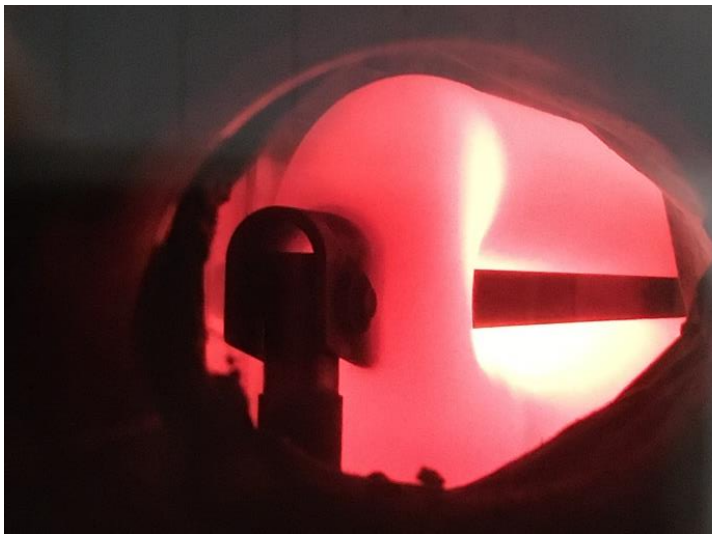
- A. Garcia-Martino, S. Fernández, J. Santiago, and M. M. Prieto, “The consequences of eliminating the soaking section of a commercial hot-dip galvanising line for HSLA production and a control strategy for improving product consistency,” Ironmaking & Steelmaking, pp. 1–7, May 2019

- “Annealing HSLA Steel without soaking phase”, A García Martino, V. Barrón, S. Fernández, J. Santiago, M. M. Prieto, Galvatech 12-16 November, Tokio
- “Experimental tests to develop an optimized water nozzle system for cooling hot galvanized strip”, S. Rodríguez, A. García Martino, B. Pereiras, Spain Minergy 17-19 June 2015, Gijón
- “Experimental study of a low-cost radiometer for hostile scenarios” G. Leon and A. García Martino, Progress In Electromagnetics Research Letters, Vol. 51, 119-125, 2015

Relevant infrastructure

Radiant tubes pilot plant.





4.1.12. R-Tech

Steinbeis Advanced Risk Technologies (R-Tech)	
www.risk-technologies.com	
Germany	*SME*

Description

Steinbeis Advanced Risk Technologies (R-Tech) is one of the transfer centers belonging to the Steinbeis Foundation (http://www.stw.de/), established in 1971. The Foundation and its 1000+ Transfer Centers in 50+ countries offer global services in technology and knowledge transfer. The Steinbeis Transfer Network provides a direct link between customers and leading experts in a variety of specialist fields, safeguarding transfer from research and development, consulting, expert evaluation, and training and education. More than 1,500 employees and 3,600 freelancers carried out the projects.

Steinbeis Advanced Risk Technologies Group founded in 2001 deals with multiple aspects of risks, risk engineering and risk management primarily in the areas of: petro chemical and process plants, power plants, material technologies, especially advanced/new material technologies, new emerging technologies (e.g. CO2, H2,nanotechnologies…). Main topics of risks dealt with are: risks in/of innovation (e.g. risks of unexpected side effects), risk of non-performance or performance below expectations (e.g. risks of system or component failures), risk of adverse/unexpected effects and impacts (e.g. on public health and/or environment), risks over the life-cycle of products and technologies (e.g. unexpected problems in decommissioning or recycling phase), project risks, especially in innovation, R&D and new technologies oriented projects.

Steinbeis R-Tech has a large experience in participating in European funded projects in the area of nanotechnology, such as the ones related to Life Cycle Assessment, Life Cycle Costing, Cost Benefit Analysis and Risk Analysis. In addition to that, within iNTeg-Risk project it was developed the CWA 16649:2013 – "Managing emerging technology-

Main tasks

In the project, R-Tech will have overall responsibility for the following work:

- Environmental and economic evaluation dealing with the activities related to life cycle assessment (LCA), life cycle costing (LCC), including the 'business intelligence' related tasks.
- Standardization activities for the new nanomaterials, devices and components.
- Standard opportunities identified in the project.
- Project risk assessment.

Relevant publications/products/services/activities

Projects:

- **MUST** - Multi-Level Protection of Material for Vehicles by "Smart" Nanocontainer. http://www.must.risk-technologies.com/
- **NANODEVICE** - Novel Concepts, Methods, and Technologies for the Production of Portable, Easy-to-Use Devices for the Measurement and Analysis of Airborne Engineered Nanoparticles in Workplace Air. http://nanodevice.eu-vri.eu/
- **MATRANS** - Micro and Nanocrystalline Functionally Graded Materials. http://kmmvin.eu/node/134
- nanoSTAIR - A Platform to Support Standardization, Innovation And Research In The Field Of Nanotechnologies. http://nanostair.eu-vri.eu
- **FIRE-RESIST** - Developing novel fire-resistant high performance composites. http://www.fire-resist.eu/FireResist/index.xhtml
- **HELM** - High-frequency electro-magnetic technologies for advanced processing of ceramic composites.
- **ExoMet** - Physical Processing of Molten Light Alloys under the influence of external fields.
- **POEMA** - Production of coatings for new efficient and clean power plants materials.

Current Projects:

- **caLIBRAte** - Performance testing, calibration and implementation of a next generation system-of-systems Risk Governance Framework for nanomaterials. http://nanocalibrate.eu/home
- **BIO4SELF** - Fibre-Based Materials for Non-Clothing Applications. http://www.bio4self.eu/
- **HARVEST** - Development of multifunctional TEG-enabled structural composite materials for the Aeronautics sector. https://www.harvest-project.eu/

Publications:

- Jovanovic, A. (2011) Life cycle Assessment (LCA) & Risk Analysis in Nanomaterials-related NMP projects. In: Proceedings for the LCA Meeting, Brussel, March 2nd 2011.
- Jovanovic A, Quintero FA. LCA and Risk Assessment for engineering nanomaterials in Europe: Going beyond conventional approaches, methods and techniques. In Workshop LCA of manufactured nanomaterials and nanotechnology based applications. Bremen, Sept. 26, 2011.
- Jovanovic A. Mapping current project profiles, research work, capacities and competences related to LCA and Risk Assessment for engineering nanomaterials in Europe. In Symposium for Safety issues of nanomaterials along their life cycle. Barcelona, May 4-5, 2011.
- Jovanovic A. Life cycle Assessment (LCA) & Risk Analysis in Nanomaterials-related NMP projects. In: Proceedings for the LCA Meeting, Brussels, March 2nd 2011.
- Breedveld L, Löscher M, Jovanovic. Outline of the iNTeg-Risk LCA guidelines: combining LCA and RA for the integrated risk management of emerging risks. In 2nd iNTeg-Risk Conference: New Technologies and Emerging Risks - Dealing with multiple and interconnected emerging risks. Stuttgart, Jun. 15-16, 2010. Stuttgart: Steinbeis. ISBN: 978-3-938062-33-3
- Jovanovic A et al. The RiskClock: A way to improve monitoring of emerging risks related to New Technologies. In 3rd iNTeg-Risk Conference: Risk vs. Risk - Managing Emerging Risk-Benefit Tradeoffs in Complex Systems. Stuttgart, Jun. 7-8, 2011. Stuttgart: Steinbeis. ISBN: 978-3-941417-65-6.

***Products / Services / activities**:*
N/A

Key personnel

Prof. Aleksandar Jovanovic, (male) is the director of the Steinbeis Advanced Risk Technologies group in Stuttgart, Germany providing consultancy in the areas of risk assessment and management for industry and public sector. As from 2006 he is also the CEO of European Virtual Institute for Integrated Risk Management (EU-VRi) and full professor at Steinbeis University. He has acted as Seconded National Expert (Germany) with the EU in Brussels, Belgium, Directorate-General Research – Industrial Technologies and Materials. His previous teaching assignments were in France (Ecole Polytechnique), Japan (University of Tokyo), USA (La Jolla) and other countries. A. Jovanovic has over 50 large international/multinational projects in the area innovation management, new technologies, business risk management, structured project management, advanced data analysis and data mining, and related areas.
Main clients in the projects have been from the EU, national governments (Norway, Belgium, Japan…), industry, utilities, insurances companies, R&D and academia. Main topics covered by the projects have dealt with risk management in industry including HSSE (Health, Safety, Security, Environment), RCM (Reliability Centered Maintenance), RBI (Risk-Based Inspection), KPIs (Key Performance Indicators) and RCFA (Root Cause Failure analysis), and many applied in industry and/or used as the basis for further standardization. A. Jovanovic is an author of five books and over 200 publications.

Dr. Parva Chhantyal, (female) graduated from the University of Manchester, England with a Master degree in Chemical Engineering in 2013. She then moved to Hannover University, Germany to pursue her PhD degree in Nanotechnology. During her PhD, she also held Research Assistant position at Laser Zentrum Hannover e.V. Her work experience and PhD specialized in understanding the optical properties of Nano-materials. Since completing her PhD in November 2017, she has been working at Steinbeis R-Tech as a Project Manager in the areas of risk and environment related issues in European and national funded projects. Her main working areas are risk analysis, life cycle assessment (LCA) and life cycle costing (LCC).

Dr. Somik Chakravarty, (male) Somik Chakravarty PhD., joined Steinbeis R-tech as a Project Manager in 2018. He holds a doctorate degree from Sorbonne Universités - UTC in Process Engineering with a specialisation in the

field of characterisation of mechanical properties on fine and cohesive bulk solids. He was awarded the Jean Bricard Prix 2019 and was nominated for the Prix de thèse Guy Deniélou 2019 for his PhD thesis. He completed his doctoral training as part of the EU FP7 Marie Curie Initial Training Network T-MAPPP working with a consortium of academic and industrial partners across Europe. He is a mechanical engineer from the University of Sheffield.

Relevant infrastructure

The SME contribution in the project will be multidisciplinary expertise on economy, market analysis, communication management, community management, IPR protection and legal advice.

4.1.13. KNEIA

KNEIA S.L.		
http://www.kneia.com		
Spain	SME	

Description

KNEIA is a Spanish SME that works with public companies, research institutions, government agencies and investors to provide assessment services on innovation management, dissemination and communication strategies, knowledge transfer, and implementation of scientific technological structures worldwide, for the improvement of the competitiveness and the technological progress of companies. Key activities include valorisation studies, the evaluation of knowledge to determine market feasibility in other sectors, the protection and valuation of R&D results, the coordination of marketing and communication activities directed at commercialising new technologies, the definition of dissemination strategies to reach potential end-users and acquirers of technologies, market analysis, trainings, eco-innovation guidelines and open innovation as well as general legal guidance on the preparation of transfer contracts.

KNEIA offers expertise in a variety of roles in project collaboration including project coordination, dissemination and communication activities, research valorization and exploitation, design management, market development and commercialization activities.

Main tasks

In the project, KNEIA is the leader of WP7. KNEIA will have overall responsibility for the dissemination related activities, including the generation of the project's business strategy as defined in the Plan for the Exploitation and Dissemination of Results (PEDR), ensuring the dissemination activities are undertaken to maximize the exploitation opportunities for the project partners and for widespread dissemination of the technical achievements demonstrated in this project.

Key personnel

Cristina Barragán has a Degree on Audiovisual Communication and is Master on International Business Administration by the Polytechnic University of Catalonia. She holds more than 15 years of experience in the area of R&D management. In relation with Research and Innovation related activities, Ms. Barragan has worked at the Open University of Catalonia, the Polytechnic University of Catalonia, the Terrassa Health Consortium and the Leitat Technological Center, where has been the Project Management Director and member of the Executive Board for several years. She has been jury member for the Technology Review TR35 Spain 2012 award, and is co-founder of KNEIA S.L.

Ciro Avolio has a Degree on Diplomatic Sciences and is Master of Environment, Development and Politics by the Sussex University. He has PhD on Sustainability at the Polytechnic University of Catalonia, with a specialization in Water Management. Has more than 10 years of experience on R&D projects management, in relation with different financing programs, including FP7, CIP, LIFE+ and ENPI amongst other. Mr. Avolio is co-founder of KNEIA S.L. and collaborates on a constant basis with international institutions, both public and private, on the definition of research and innovation projects, focused on the assessment of the valorisation of their projects portfolio.

Relevant publications/products/services/activities

KNEIA is a young company, created by a senior team with long experience in Collaborative Programmes, including the development and implementation of innovation business plans for SMEs, supporting access to R&D funding opportunities at European level (H2020, FP7, LIFE+ and CIP programmes among others), developing ad-hoc dissemination and communication strategies, conducting market research analyses and technology monitoring assignments, delivering specialist training sessions on the adoption of innovative practices, and defining route-to-market and proof-of-concept operations on new product development and launch.

Previous projects:
KNEIA personnel has wide experience on leading the Work Package on Dissemination and Exploitation of R+D results in H2020 and FP7 projects, offering a vast experience covering following contributions:

- Dissemination and Communication of R&D results: designing suitable communication strategies and campaigns and supporting the whole dissemination process
- Project & Risk Management: valuating project risk, defining project governance and management best practice, and providing technical assistance and project management support to collaborative projects.
- IP Strategy Design: Protection and valuation of intangible assets generated from R&D projects, and general assistance on regulatory and legal issues related to technology.
- Commercialization of R&D results (Knowledge Marketing): Supporting the technology transfer process, through the evaluation, marketing and deal-making of technology.

Our main projects are:

- H2020 **ECOBULK -** Circular Process for Eco-Designed Bulky Products and Internal Car Parts
- H2020 **ZEOCAT-3D -** Development of a bifunctional hierarchically structured zeolite based nano-catalyst using 3D-technology for direct conversion of methane into aromatic hydrocarbons via methane dehydroaromatization
- H2020 **ECOAT -** ECO sustainable multifunctional biobased COATings with enhanced performance and end of life options

Relevant infrastructure
No infrastructures will be made available by KNEIA. The SME contribution in the project will be multidisciplinary expertise on economy, market analysis, communication management, community management, dissemination, IPR protection and legal advice.

4.2. Third parties involved in the project (including use of third party resources)

4.2.1. UNIPI

No third parties involved.

4.2.2. IPCF-CNR

No third parties involved.

4.2.3. FM

No third parties involved.

4.2.4. FHF

No third parties involved.

4.2.5. UOB

Does the participant plan to subcontract certain tasks (please note that core tasks of the project should not be sub-contracted)	Y
If yes, please describe and justify the tasks to be subcontracted 5.925,00 € Characterisation costs that cannot be done at UoB 2.000,00 € IP protection costs	
Does the participant envisage that part of its work is performed by linked third parties[1]	N
If yes, please describe the third party, the link of the participant to the third party, and describe and justify the foreseen tasks to be performed by the third party	
Does the participant envisage the use of contributions in kind provided by third parties (Articles 11 and 12 of the General Model Grant Agreement)	N
If yes, please describe the third party and their contributions	
Does the participant envisage that part of the work is performed by International Partners[2] (Article 14a of the General Model Grant Agreement)?	N
If yes, please describe the International Partner(s) and their contributions	

4.2.6. Archer Technicoat Ltd

No third parties involved.

[1] A third party that is an affiliated entity or has a legal link to a participant implying a collaboration not limited to the action. (Article 14 of the Model Grant Agreement).

[2] 'International Partner' is any legal entity established in a non-associated third country which is not eligible for funding under Article 10 of the Rules for Participation Regulation No 1290/2013.

4.2.7. ENEA

Does the participant plan to subcontract certain tasks (please note that core tasks of the project should not be sub-contracted)	N
If yes, please describe and justify the tasks to be subcontracted	
Does the participant envisage that part of its work is performed by linked third parties[3]	Y
If yes, please describe the third party, the link of the participant to the third party, and describe and justify the foreseen tasks to be performed by the third party CertiMaC will perform a thorough thermo-physical characterization of the solutions developed by the other Partners. CertiMaC will assess thermal conductivity and specific heat by means of both traditional experimental methods (heat flow meter) and innovative methods (Laser Flash Analysis - LFA method) up to 1250°C. CertiMaC will aim at estimating the thermal efficiency of the designed solutions by simulating the behaviour of the materials in real working conditions. The results of the simulations will be employed for the evaluation of heat transfer ability and for the improvement of the thermal efficiency of the materials. The budget allocated to the Third party for the execution of its activities is estimated to be 60.000 euro.	
Does the participant envisage the use of contributions in kind provided by third parties (Articles 11 and 12 of the General Model Grant Agreement)	N
If yes, please describe the third party and their contributions	
Does the participant envisage that part of the work is performed by International Partners[4] (Article 14a of the General Model Grant Agreement)?	N
If yes, please describe the International Partner(s) and their contributions	

4.2.8. CNRS

Does the participant plan to subcontract certain tasks (please note that core tasks of the project should not be sub-contracted)	N
If yes, please describe and justify the tasks to be subcontracted	
Does the participant envisage that part of its work is performed by linked third parties[5]	Y
If yes, please describe the third party, the link of the participant to the third party, and describe and justify the foreseen tasks to be performed by the third party The following parties will also participate in the project as third party contractually linked with CNRS: Université de Bordeaux and CEA through the joint research unit n°5801, also called LCTS - Polytech Orléans and Université d'Orléans through the joint research unit n°3079, also called CEMHTI - Université de Technologie de Belfort-Montbéliard through the joint research unit n°6303, also called ICB. This involvement is due to the fact that staff involved in CEM-WAVE are employed by these institutions. Contracts have been signed between CNRS and these institutions regarding the JRUs ("Quinquennal contract" of a five-year period or equivalent) stating the resources (human, financial, infrastructures...) allocated by each institution to each laboratory for research purposes." The budget allocated to the Third parties for the execution of their activities is estimated to be 118.736 euro.	
Does the participant envisage the use of contributions in kind provided by third parties (Articles 11 and 12 of the General Model Grant Agreement)	N

[3] A third party that is an affiliated entity or has a legal link to a participant implying a collaboration not limited to the action. (Article 14 of the Model Grant Agreement).

[4] 'International Partner' is any legal entity established in a non-associated third country which is not eligible for funding under Article 10 of the Rules for Participation Regulation No 1290/2013.

[5] A third party that is an affiliated entity or has a legal link to a participant implying a collaboration not limited to the action. (Article 14 of the Model Grant Agreement).

If yes, please describe the third party and their contributions	
Does the participant envisage that part of the work is performed by International Partners[6] (Article 14a of the General Model Grant Agreement)?	N
If yes, please describe the International Partner(s) and their contributions	

4.2.9. POLITO

No third parties involved.

4.2.10. IRIS

No third parties involved.

4.2.11. AM

Does the participant plan to subcontract certain tasks (please note that core tasks of the project should not be sub-contracted)	Y
If yes, please describe and justify the tasks to be subcontracted 40.000 € → Support during the preparation and execution of the trials: adaptation works, monitoring of the trials, resolution of technical incidences	
Does the participant envisage that part of its work is performed by linked third parties[7]	N
If yes, please describe the third party, the link of the participant to the third party, and describe and justify the foreseen tasks to be performed by the third party	
Does the participant envisage the use of contributions in kind provided by third parties (Articles 11 and 12 of the General Model Grant Agreement)	N
If yes, please describe the third party and their contributions	
Does the participant envisage that part of the work is performed by International Partners[8] (Article 14a of the General Model Grant Agreement)?	N
If yes, please describe the International Partner(s) and their contributions	

4.2.12. R-Tech

No third parties involved.

4.2.13. KNEIA

No third parties involved.

[6] 'International Partner' is any legal entity established in a non-associated third country which is not eligible for funding under Article 10 of the Rules for Participation Regulation No 1290/2013.
[7] A third party that is an affiliated entity or has a legal link to a participant implying a collaboration not limited to the action. (Article 14 of the Model Grant Agreement).
[8] 'International Partner' is any legal entity established in a non-associated third country which is not eligible for funding under Article 10 of the Rules for Participation Regulation No 1290/2013.

4.3 Third parties' description

4.3.1 Certimac

CertiMaC Soc. Cons. a R. L. (CERTIMAC)	
https://www.certimac.it/Default.cfm?lingua=EN	
Italy	*Private no-profit Research Centre*
Role: Third Linked Party of ENEA	

CertiMaC is a Private no-profit Research and Technology Transfer laboratory for Innovation in Materials science and energy sector, with a specific focus on Buildings & Construction. Founded by ENEA (National Agency for Energy Efficiency) and CNR (National Research Council) – the two main actors of scientific research in Italy, CertiMaC offers a wide range of services related to R&D, characterisation and certification for traditional and innovative materials and components. In particular, CertiMaC provides services and research activity aimed at the assessment and engineering of mechanical, chemical, physical and thermo-hygrometric performances:

- certification - CE Marking, Compliance and Quality Certificates;
- research and consultancy - normative and pre-normative research on materials and innovative solutions at EU and national level;
- development of research projects, new methods of testing and technology transfer initiatives;
- consulting and testing services, which offer material manufacturers, building professionals the most advanced solutions available within National research, particularly in the fields of Energy Efficiency and materials science.

CertiMaC is also recognised by the European Commission and by the Italian Ministry of Infrastructure and Transport as Notified Body no. 2685 in accordance with the Construction Product Regulation - CPR (EU) 305/'11. CertiMaC works under ISO/IEC 17025 and is certified in accordance with ISO 9001:2015 Management System.
At national level, CertiMaC is also accredited within the High Technology Network of the Emilia-Romagna Region and is funding member of two Technological Clusters - Build (for Construction sector) and Green&Tech (for Energy & Environment) - aimed at meeting the needs of enterprises by providing expertise, tools and solutions.
CertiMaC is also member of UNI (National Standardisation Body) and of ECTP (European Construction Technology Platform).

R&D Units: Energy Efficiency and Material Science Units are the two main units of CertiMaC. They are focused on thermo-physics performance of materials and components not only in order to develop new standards, solutions, materials but also to provide know-how and skills to the Industry.

Main tasks and responsibilities

The expertise acquired thanks to many years' activity in the field of thermo-physical characterization will allow CertiMaC to give a significant contribution to WP4. CertiMaC will perform a thorough thermo-physical characterization of the solutions developed by the other Partners. CertiMaC will assess thermal conductivity and specific heat by means of both traditional experimental methods (heat flow meter) and innovative methods (Laser Flash Analysis - LFA method) up to 1250°C. The combination between thermal conductivity and specific heat will enable CertiMaC to determine a fundamental parameter - thermal diffusivity (mm2/s) - which plays a strategic role in qualifying component/material behaviour in the framework of thermal exchange problems.

Moreover, CertiMaC will aim at estimating the thermal efficiency of the designed solutions by simulating the behaviour of the materials in real working conditions. The results of the simulations will be employed for the evaluation of heat transfer ability and for the improvement of the thermal efficiency of the materials.

Relevant publications/products/services/activities

CertiMaC has developed a large number of Projects and Publications in the field of thermo-physical characterization and qualification of materials and components. The most relevant projects (related to the proposal) and publications are listed below:

Relevant Projects

- **Horizon2020 – Nextower: Role: Partner - Year: 2017-2020** - NEXTOWER shall introduce a set of innovative materials to boost the performance of atmospheric airbased concentrated solar power (CSP) systems to make them commercially viable. In particular, tower systems are appealing for the great environmental compatibility and offer tremendous potential for efficient (electrical and thermal) power generation. CertiMaC took part in proposal writing and is managing WP4 about the experimental campaigns devoted to the energetic characterisation of metallic and ceramic materials developed.
- **ERDF Regional Project (Emilia Romagna) - EEE-CFCC - Year 2016-2018.** The project EEE-CFCC involves the realization of a new thermostructural and basalt fiber reinforced composite material (BasKer) for high temperature applications for transport and construction sector. CertiMaC characterized the materials and the solutions developed within the Project and carried out the simulation of the behavior of the materials under the working conditions.
- **ERDF Regional Project (Emilia Romagna) - FIREMAT – Year 2019-2021-** FIREMAT is dedicated to the development of low-density thermostructural materials for lightening and increasing safety, specifically long and short fiber reinforced composites, for high temperature and fireproof applications. CertiMaC, as part of the FireMat project, deals with the activities of industrial research and experimental development aimed at engineering materials and components for high temperature applications, identifying the most relevant thermo-physical characteristics such as thermal conductivity, specific heat and diffusivity, and studying their reaction to fire behavior. Both areas of study will also be expanded and deepened by a numerical modeling stage dedicated to studying the behavior in operation of the various systems developed.

***Relevant Publications**:*

- E. Papa, V. Medri, An Natali Murri, L.Laghi, G. De Aloysio, S. Bandini E. Landi, Characterization of alkali bonded expanded perlite, Construction and Building Materials, 191, (2018), 1139-1147
- Morganti M., Lignola G.P., Caggegi C., Ceroni F., De Santis S., Krajewski P., Lourenço P.B., Papanicolaou C., Pellegrino C., Prota A., Zuccarino L.Performance assessment of basalt FRCM for retrofit applications on masonry. (2017). Composites Part B: Engineering.
- G. De Aloysio, G. D'Alessandro, F. de Monte, (2016), An analytical solution for the hyperbolic unsteady thermal behaviour of micro-thermoelectric coolers with a suddenly time-dependent heat generation, International Journal of Heat and Mass Transfer,95, 972-983;
- L. Laghi, F. Pennecchi and G. Raiteri (2011). Uncertainty analysis of thermal conductivity measurements in materials for energy-efficient buildings. International Journal of Metrology and Quality Engineering, 2, pp 141-151;

Key personnel

Luca Laghi (Male) - Mechanical Engineer and Technical Director. Head of R & D and Energy Efficiency Unit focused on materials' thermo-hygrometric and thermo-physical properties improvement and FEM models simulating heat transfer through materials and components, industrial applications and insulation issues in steady and unsteady state transfer. He is enrolled in the Engineer Register of Ravenna and in the Register of Energy Certifies. Publications in national and international journals for what concerns studies on Thermal Conductivity measurements and analysis. Expertise in European and national legislation in Energy Efficiency in buildings, Energy Certification of Buildings, Energy Management. Project manager into regional and national Level coordinated directly by CertiMaC and Project Manager in several EU Projects for the partner CertiMaC as WP Leader. Trainer in Higher Education Courses on: Retrofit of existing buildings and diagnostic analysis of Building envelope and technical systems, Innovative Materials and Solutions for energy efficient and sustainable buildings.

Giulia De Aloysio (Female) - PhD in Systems and Methods for Management of electric and thermal energy from renewable and assimilated sources and for the sustainable building (University of L'Aquila, ITALY, 2016). Researcher at CertiMaC. She previously worked as scientific collaborator at Fraunhofer Institut fuer Bauphysik (IBP) – Stuttgart, Germany, in the field of building physics focused on thermal conductivity measurements with steady state and transient methods. Her research fields are thermos-physical characterization of building materials and

energy efficiency in buildings. Publications in international Journals for what concerns the thermal characterization of micro thermoelectric coolers.

Elena Casiraghi (Female) - Master Degree in Business Administration. Interpreter and Translator: RTD Grant writing and Senior Project Manager of European projects. Expert in Project Monitoring methodologies and strategies. Expertise in management of international relations and international working groups.

Infrastructure

Equipment and Facilities for Experimental Analysis and campaigns relevant for this proposal are listed below:

- 1 guarded heat flow meter to determine thermal conductivity and hygro-thermal performance of envelope materials within a range of temperature of -20 to 300 °C devoted to material with low-medium thermal resistance;
- 1 heat flow meter to determine thermal conductivity and hygro-thermal performance of envelope materials within a range of temperature of 0 to 100 °C devoted to material with medium-high thermal resistance;
- 1 Laser Flash Analyzer able to determine thermos-physical properties of a wide range of materials up to 1250 °C through the determination of Thermal Diffusivity and, indirectly, Specific Heat and Thermal Conductivity;
- 1 DSC able to determine specific heat, glass transition temperature and other phenomena such as Sol-liq fusion - Crystallization - Evaporation - Condensation - Sublimation - brination - Chemical reactions starting from -90 °C and up to 700°C.
- 1 IR professional Thermocamera operating between -50 and 1200 °C to evaluate temperature distribution into existing civil and industrial structures/processes and to diagnose issues related to use and performance (e.g. degradation, thermal bridges, etc.);
- 1 Portable Wi-Fi heat flow meter to evaluate on field thermal transmittance and Conductance of existing walls and partitions with the main aim to determine the energy performance “as it is” and to evaluate energy improvements.
- fully automated Climatic Chambers (0.7 x 0.9 x 2 m) for weathering resistance determination within the range -30 ÷ 80 °C. Climate simulation by means of air, water, water spray, de-icing salts, salt fog, UV, etc.;

Section 5: Ethics and Security

5. Ethics Issues

5.1 Ethics

No ethic issues need to be considered in this proposal, according with the Participants Portal Ethics form.

5.2 Security

Please indicate if your project will involve:

- activities or results raising security issues: **NO**
- 'EU-classified information' as background or results: **NO**

SACMI

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Società appartenente al Gruppo Sacmi Imola S.C.

Salvaterra, 31.01.20

Dear Prof. Andrea Lazzeri
Coordinator of proposal CEM-WAVE, within call H2020 LC-SPIRE-O8-2020

SACMI has been informed by Claudio Mingazzini of ENEA about the proposal you are submitting, aiming at producing industrial kiln, with lower C-footprint. As SACMI we are very interested in your approach (direct burning of hydrogen produced by renewable energies, or hydrogen plus bio methane mixtures) requiring the development of innovative ceramic composite materials for burners and radiant tubes. We consider this program much in line with our strategic priorities, as reflected by our participation to ASPIRE association.

We would like to be kept informed about CEM-WAVE proposal outcomes, and also welcome the future possibility of joining an International Advisory Board of Stakeholders.

Your Sincerely,

SACMI FORNI CEO
Stefano Lanzoni